Method-Evolution Mechanics: From Experience to Method in LLM Research Systems

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Abstract

Long-running large-language-model research agents produce more than candidate answers. They generate longitudinal evidence about how research is conducted: which representations are chosen, which prior experiences are retrieved or missed, which methods are repeated, where local results fail to connect to a root problem, and how verification or provenance defects alter the trajectory. Existing work has shown that agents can learn from episodic memory, induce reusable workflows and skills, and modify their own code. We ask a stricter question: *can the research method itself evolve from these trajectories under an evidence regime that prevents the system from certifying its own improvement?*

We study this question in Orion, a persistent external research substrate in which immutable task episodes, versioned lessons, failures, operators, strategy motifs, problem-conditioned fibres, gluing obstructions and meta-method variants are overlapping typed views. Recent framework hardening makes several process boundaries executable: proposal-shadow episode storage is distinct from canonical admission; consequential learning-turn execution can require a content-bound pre-action receipt so retrospective traces do not receive prospective credit; cross-problem "no relevant memory" claims require a bounded coverage receipt; and an epistemic-noninterference checker attacks ways in which experience-side state could be confused with scientific authority. The deployed mathematical-research path contains a content-bound obstruction–transformation memory whose structural episodes feed invention-last `SEARCH`, witnessed `JUMP`, interface-checked `GLUE` and residual/coverage-bound `LIFT`. A successful target transformation may later be consolidated as a scoped structural episode and, separately, a reusable tool; neither occurs automatically from one success. The current runtime composes an immutable scientific-authority projection, but experience-side learning remains non-sovereign: storing, retrieving or promoting a method cannot by itself promote a scientific claim.

We use a live multi-agent programme spanning six unsolved Millennium Prize Problems as a longitudinal case study and define a seven-axis retained-growth vector, typed process telemetry, experience-to-lesson/tool conversion, repeated-failure measures and a four-arm causal-attribution benchmark comparing `MODEL_ONLY`, `RAKL_RESET`, budget-matched `RAKL_SHAM_MEMORY` and persistent `RAKL_LEARNING`. A two-arm precursor experiment remains immutable negative history: under Qwen2.5-0.5B-Instruct both arms produced zero registered successes. The later 0.5B–7B ladder also remains immutable as the result of the then-frozen instrument. Subsequent root-cause audit, however, found that the sealed evidence-binding interface did not define the semantic roles of its selected/rejected evidence-ID fields; a hosted GLM-5.2 probe then cleared the conceptual capability floor while saturating the old single-task panel. Those later results do not retroactively turn the old negative positive. They localize the unresolved model-adapter problem to a fresh, discriminating evidence-binding qualification rather than making model capability a required dependency of the method-state architecture.

The required method mechanics are tested independently of that adapter residual. The production routing stack binds positive `SEARCH/JUMP/GLUE/LIFT/CANNOT_CHECK` routes to the canonical audit, exact candidate/revision/target-bound rejection certificates to conclusive

SEARCH/JUMP failures, exact composition/order/revision/content-bound certificates to conclusive GLUE failures, and post-success consolidation to a separately verified target episode; missing, stale, ambiguous or non-conclusive evidence remains `CANNOT_CHECK`, and these behaviors are enforced by exact-head conformance tests. An earlier quoted 21-family, 2,688-case adversarial validation of this router is deliberately not cited as evidence in this paper: its frozen protocol survives, but no harness, case set or result artifact for it exists in the repository, so the quoted numbers are unreproducible and carry `CANNOT_CHECK` status until the harness is produced or the result is withdrawn.

A second frozen successor isolates the experience-to-action dependency itself once experience has been verified and structured. The load-bearing result is an *information-sufficiency* separation, stated as such: on a frozen 1,792-case, 14-family benchmark the registered typed experience coordinates determine the governed action, while the union of the registered reset, failure-memory, scalar-confidence, provenance and untyped whole-state parents has deterministic information ceiling 0.714, unsafe-apply rate 0.286 and zero `CANNOT_CHECK` recall; all 10 registered hostile mutations are detected. A governance audit found that this instrument's four accuracy-shaped gate conditions were satisfied by construction and could not fail; they are preserved as narrowed instrument history, not reported as candidate measurements. A repaired successor gate with an independently implemented declarative oracle and post-freeze gold then measures the shipped mechanic on a fresh 2,400-case panel: exact conformance (1.000 accuracy, 0 unsafe applies, 1.000 `CANNOT_CHECK` recall), a composite-parent information ceiling of 0.433, 12/12 planted candidate mutants caught, and a per-condition black-box falsifiability audit returning FALSIFIABLE on all eight evidence-dependent gate conditions with controls asserted first. This supports the *structured verified experience-to-action transition* as the required method mechanic at implementation-conformance strength. It does not claim that an arbitrary language model can infer those structured lessons from raw text, and the four-arm fresh-task lift remains a prospective, capability-qualification-blocked measurement, not a null.

We further specify a governed upgrade protocol for Orion 3.x in which challengers are content-identified, fresh transfer is protected from proposer access, evaluators and thresholds are protected within an evaluation epoch, previous variants remain rollback targets and post-promotion attestation is separated from pre-promotion authorization. The paper therefore supports an evidence-governed method-evolution architecture whose required structured experience, routing, rejection and consolidation mechanics are promotion-grade, while preserving historical model-interface negatives as non-sovereign adapter evidence. The Phase-0 retained-novelty audit substrate is retained as internal metrology only because the independent track contained zero external novelty labels at freeze. No claim is made that Orion universally improves fresh LLM research outcomes, establishes independent semantic novelty, solves any Millennium problem, or grants scientific authority from method-state evolution itself.

1 Research systems should learn how to research

The standard unit of evaluation for an AI agent is a task outcome. A task is given, the agent acts, and the system is scored on success, correctness, reward or cost. This abstraction is appropriate for many deployed agents, but it discards much of the evidence produced by open-ended research. A research run contains a sequence of representation choices, decompositions, literature searches, retrieved analogies, rejected tools, conjectures, falsifiers, source checks, local proofs, failed bridges and residual questions. Even when the root problem remains unsolved, these intermediate events reveal which research methods were useful, which failed, and why.

This distinction matters because long-horizon autonomous research is not well described by repeated independent benchmark queries. A recent behavioural case study of roughly one hundred sequential architecture experiments found a rapid-gain phase, an extended saturation wall and later recovery after the action surface was expanded; it also traced part of the observed greedy search behaviour to the workflow itself rather than to the underlying model alone [1]. The result suggests that the research harness is not merely an implementation detail. It can shape what hypotheses are generated, how risk is taken and when the system escapes a local research regime.

At the same time, a growing family of agent systems learns from externalized experience. Reflexion stores verbal reflections [2]; ExpeL extracts reusable insights from collections of trajectories [3]; Voyager accumulates an executable skill library [4]; Agent Workflow Memory induces reusable routines [5]; and more recent systems learn hierarchical skills or dual streams of experience and skill knowledge [6, 7]. These systems establish that persistent non-parametric experience can improve later behaviour. They also expose a new control problem. External memory can create negative transfer, retrieval competition and consolidation failure. In sequential settings, the continual-learning bottleneck may move from model weights to memory representation and access [8]; repeated LLM consolidation can even corrupt useful experience, motivating raw episodic preservation as a first-class design requirement [9].

A separate line of work asks agents to improve the agent architecture itself. Automated Design of Agentic Systems searches over agent programs [13]; Gödel Agent and related systems permit self-referential policy modification [14]; and Darwin Gödel Machine maintains an open-ended archive of self-modifying coding agents selected through empirical evaluation [15]. These approaches make recursive improvement operational. They also sharpen an unanswered engineering question for research systems: *what evidence licenses the claim that a self-modification is an improvement when the same system can propose the modification, influence the evaluator, select the tasks and rewrite the state from which future evidence is interpreted?*

Our previous ORION work addressed a different layer of this problem. Paper 4 introduced a verified-discovery architecture for LLM-mediated mathematical research in which specification alignment, theorem truth, novelty, research value and verifier trust are separate authority coordinates; proposal generation cannot promote itself to theorem authority; and computation, proof and novelty are kept distinct [22]. The present work asks the next question. Once research trajectories are preserved under such an authority regime, can they be used to improve the *method of research itself* without collapsing observed performance, deployment status and epistemic authority into one scalar notion of "better"?

1.1 Central claim and evidence boundary

Our central claim is architectural and methodological:

> A long-running LLM research system can treat its own research trajectories as typed evidence for method improvement if raw episodes are preserved, diagnoses and abstractions remain defeasible, method changes are evaluated as content-identified challengers against frozen matched baselines and fresh transfer tasks, and promotion is separated from both proposal generation and operational deployment.

Figure 1 shows the loop this claim describes. This paper supports that claim at three different evidence levels that must not be conflated.

1. **Implemented architecture.** ORION contains executable objects for immutable task episodes, versioned lessons, failure revisions, problem fibres, local-to-global gluing, experience-conditioned routing, vector saturation, problem-novelty classification, branching evolution variants and matched experience benchmarks.
2. **Retrospective/live case-study evidence.** A multi-agent mathematical research programme in the separate `RAKL_math` repository has already produced auditable examples of retrieval misses, representation failures, source-boundary failures, local-to-global interface gaps, chronology repairs and assurance defects. These observations motivate hypotheses about research-process weaknesses. Because much of the programme predates v3 telemetry, we treat these observations as qualitative and partly retrospective.

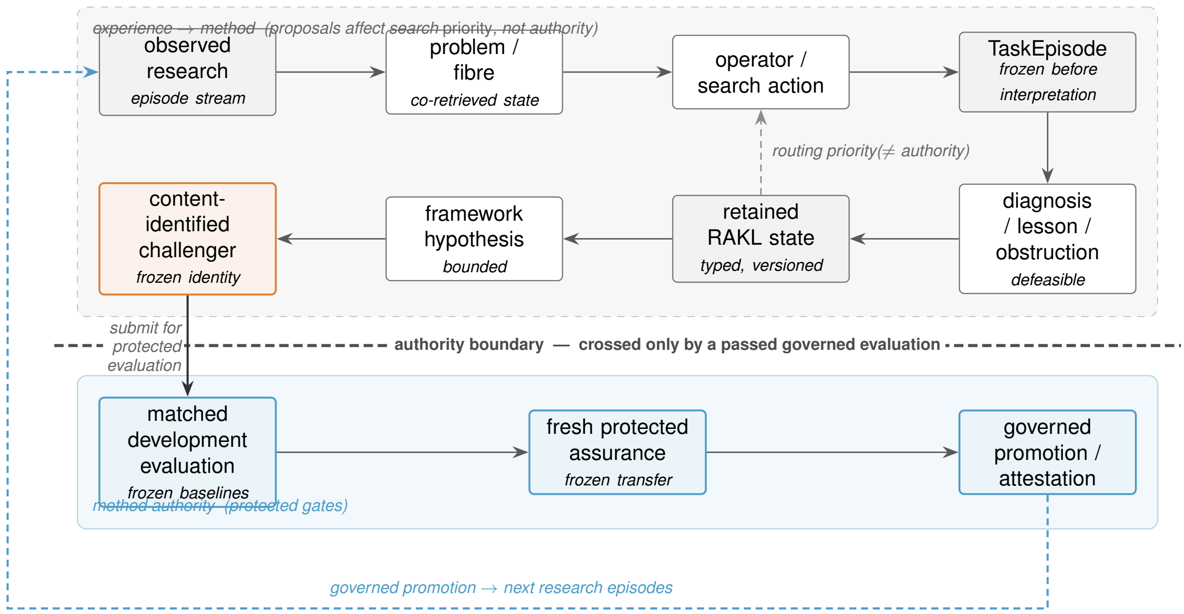

Figure 1: The experience-to-method loop. Observed research produces frozen `TaskEpisode`s; diagnoses and lessons accumulate as defeasible, versioned state that re-ranks routing priority but carries no authority. A framework change is packaged as a content-identified challenger which may cross the *authority boundary* only by passing a governed evaluation: matched development baselines, fresh protected assurance, and governed promotion. Retained experience thus affects *search priority*, never scientific or method authority directly — the same proposal/commitment separation as Paper I, applied to the method itself.

3. **Prospective claims not yet licensed.** We preregister how later ORION variants such as 3.1 and 3.2 should be evaluated. Until those matched development and fresh-assurance trials are run, we do not claim a measured general improvement in research capability, efficiency or scientific discovery rate.

The distinction is deliberate. A paper about recursive self-improvement is especially vulnerable to circular evidence. If the architecture is allowed to define success after seeing the outcome, or if a framework version is called improved merely because it was merged, the empirical claim becomes unfalsifiable.

1.2 Research as four coupled loops

We model the persistent research system as a state R_t coupled to a proposal generator with parameters θ. For problem input P_t,

$$(S_t, \tau_t) = \mathrm{Driver}_\theta(P_t, R_t), \qquad R_{t+1} = \mathrm{Learn}(R_t, \tau_t),$$

where S_t is the task output and τ_t is the externally recorded public research trajectory. The base model parameters may remain fixed while behaviour changes through the evolving external state.

The system is organized around four coupled loops:

$$\text{information} \to \text{knowledge},$$

$$\text{problem} \to \text{solution},$$

$$\text{experience} \to \text{method},$$

ORION → better candidate ORION.

The first two loops are conventional research operations. The third turns task experience into reusable but bounded method knowledge. The fourth is a meta-loop that proposes changes to the research architecture itself. Paper 5 focuses on the boundary between the third and fourth loops.

1.3 Why the unit of analysis cannot be "success" alone

Open research has sparse and delayed terminal rewards. A Millennium Prize Problem may remain unsolved for the duration of the study, while a run can still produce valuable information by refuting a method family, identifying an exact missing theorem, recovering a relevant prior result, or proving a scoped local lemma. Conversely, a locally correct lemma may create little root progress if the interface needed to connect it to the target remains exactly the hard part.

We therefore treat research progress as a vector of outcomes rather than a single score. Relevant coordinates include valid residual contraction, repeated-failure avoidance, retrieval coverage, correct rejection of unsafe transfer, proof or falsifier quality, detection of gluing/interface failure, resource use and authority preservation. A framework upgrade is admissible only if soft gains do not purchase violations of hard authority invariants.

This leads to the core methodological separation used throughout the paper:

$$\boxed{\text{DEPLOYMENT} \neq \text{PERFORMANCE EVIDENCE} \neq \text{METHOD AUTHORITY}}.$$

Code may be merged. A benchmark may improve. A method may even receive fresh assurance. These are different facts and are represented separately.

2 A longitudinal observatory for machine research

2.1 The live mathematical case study

The observational substrate is a live research programme in `SzeChunYiu/RAKL_math`. The repository is deliberately separated from the reusable ORION framework repository. The application repository contains problem contracts, context fibres, research DAGs, source packets, candidate artifacts, falsifiers, reviews, traces and problem-specific tests for six unsolved Millennium Prize Problems: P versus NP, the Riemann Hypothesis, Navier–Stokes existence and smoothness, Yang–Mills existence and mass gap, the Hodge Conjecture, and the Birch and Swinnerton–Dyer Conjecture. A cross-problem lane studies transfer and recurring process failures across these domains.

At the initial Paper 5 freeze, the active scheduled programme contains nine hourly research lanes: P versus NP; an analytic Riemann-Hypothesis lane; Navier–Stokes regularity and blow-up lanes; constructive and spectral/RG Yang–Mills lanes; a Hodge deformation lane; an analytic BSD lane; and a cross-Millennium transfer/meta-observer lane. Complementary spectral RH, motive-oriented Hodge and Selmer/Iwasawa BSD lanes are preserved but paused. The schedule is an operational fact, not a claim of nine independent replicates. All lanes share repository state and can influence one another through persistent memory, so statistical analyses must treat them as coupled longitudinal processes.

Every application run is instructed to read current ORION `main` as the framework source of truth, while new mathematical application work remains in `RAKL_math`. This split serves two purposes. First, problem evidence does not silently become framework authority. Second, a framework defect exposed by one domain can be opened as a separate ORION issue and evaluated as a method hypothesis rather than patched inside the mathematical result that exposed it.

2.2 Observation is not hidden reasoning disclosure

The observatory does not attempt to reconstruct private model chain-of-thought. The recorded object is an auditable scientific decision trace: problem state, retrieved evidence, proposed action, executed tools, observable result, verification, residual and next action. This distinction is methodologically important. A reproducible research record need not contain the inaccessible or potentially unstable private reasoning process that generated a proposal.

Definition 1 (Task episode)**.** *A task episode is an immutable externally observable record*

$$E = (p, a, c, F, O, \tau, V, y, r, \kappa, t, h),$$

where p is the problem/atom identity, a the active atomic target, c the context identity, F the problem-fibre snapshot, O the operators or methods attempted, τ the public action/observation trace, V the verification evidence, y the outcome, r the residual signature, κ a resource/cost record, t the timestamp and h the content identity.

A non-success outcome should carry an explicit residual or blocker. The raw episode is never replaced by a later reflection summary.

2.3 Episode, diagnosis and lesson are different objects

A central design decision is

$$\boxed{\text{episode} \neq \text{diagnosis} \neq \text{obstruction} \neq \text{lesson}}.$$

An episode records what happened. A diagnosis proposes why it happened. An obstruction generalizes a supported boundary that may apply elsewhere. A lesson proposes a reusable action under explicit triggers and scope conditions. Conflating these stages causes a familiar failure in autonomous agents: a fluent post-hoc explanation is treated as a verified causal rule.

A versioned lesson therefore records

$$L = (g, c, a, e, b, S^+, S^-, f, \alpha, \nu, h_L),$$

with trigger g, scope c, proposed action a, expected effects e, boundaries b, supporting and contradicting episodes S^+, S^-, a falsifier f, authority α, validation obligations ν and content identity h_L. Reusable promotion requires evidence beyond reflection. Raw episodes remain evidence roots even when a lesson becomes useful.

2.4 Problem-conditioned fibres

A research agent should not retrieve the entire archive for every action. For active problem atom A under context c, Orion compiles a problem-conditioned fibre

$$\mathcal{F}(A \mid P, c) = (K_A, O_A, E_A, B_A, M_A, X_A),$$

where K_A contains epistemic items, O_A applicable tools/operators, E_A analogous episodes, B_A failures and boundary warnings, M_A strategy motifs, and X_A expertise or unresolved warnings. Co-retrieval does not imply compatibility or authority.

The fibre snapshot is important for retrospective method analysis. If a relevant item existed in the bound search universe but was not retrieved, the failure is different from a case in which the item was not present. If the item was retrieved but rejected, the rejection rationale becomes part of the observable decision record.

2.5 Local results and gluing failures

Many difficult research episodes do not fail inside a local lemma. They fail because locally valid pieces cannot be assembled into a root result. ORION represents this explicitly. A decomposition produces problem atoms with dependencies and optional interface keys. Local sections can be verified independently, but a global solution requires complete coverage, dependency closure and compatibility on shared interfaces.

We therefore distinguish

local mathematical failure

from

gluing/interface failure.

The latter includes missing uniformity across scales, a theorem whose hypotheses are not inherited by the target limit, a representation that does not preserve a root-critical coordinate, or a proof component that is true only in a different category. Failed gluing is itself an experience record and can become a recurring obstruction family.

2.6 Vector saturation

Repeated unsuccessful search should not be summarized as "stuck." The v3 saturation state tracks bounded novelty across seven coordinates:

$$\mathrm{Sat}(R, D) = (S_K, S_O, S_E, S_B, S_R, S_P, S_M),$$

corresponding to knowledge, operator, experience-pattern, obstruction, relation, path and meta-method novelty. An axis is considered bounded-flat only under a registered search window with sufficient independent route coverage, no retained novelty on that axis and no native residual that reopens it. Saturation never licenses an absolute completeness claim.

The observatory records which axis was reopened after a consequential episode. This permits later questions such as whether an agent wastes cycles by continuing to search the knowledge axis when the stable residual is representational or relational.

2.7 Structural novelty of a solved subproblem

A verified local result can be classified by the strongest genuinely new problem-solving structure required:

Class	Interpretation
`STORED`	exact verified solution already registered
`RAKL_TRIVIAL`	composition of existing resources/operators
`TRANSFER_NOVEL`	new application of existing structure with mapping witness
`REPRESENTATION_NOVEL`	new representation/bridge/invariant required
`OPERATOR_NOVEL`	genuinely new transformation/operator required
`ONTOLOGY_NOVEL`	schema/ontology extension required
`UNRESOLVED`	verification or ancestry insufficient

This classification supports an empirical question that cannot be answered from final problem labels alone: what fraction of useful research progress requires genuinely new problem-solving primitives, and what fraction is retrieval, transfer, recomposition, representation repair or gluing?

2.8 Method-case telemetry

For Paper 5, each new consequential research cycle is instructed to emit a bounded `RAKL_METHOD_CASE_STUDY` record. The record contains:

- the research method or operator sequence actually used;
- which prior successes and failures affected routing;
- relevant items retrieved but rejected, and relevant items later discovered to have been missed;
- the chosen falsifier or verification action;
- outcome and residual;
- whether the failure is primarily mathematical, retrieval, decomposition, representation, gluing, source/provenance, verification, tooling/CI or meta-policy;
- the saturation coordinate reopened;
- the strongest defensible novelty class;
- a framework-improvement hypothesis, if one is warranted, with a cheapest prospective discriminator.

The method-case record is an observational layer. It does not authorize a framework change.

2.9 Human and infrastructure interventions

The longitudinal study also records exogenous interventions. Examples include manual schedule-prompt changes, human merge overrides, model/tool upgrades, CI changes and repository migrations. Without an intervention ledger, a later performance change can be falsely attributed to experience learning when it actually came from a stronger base model or a changed tool surface.

Accordingly, a version comparison should bind at least model identity/revision, system-prompt or harness identity, tool policy, resource ceilings, repository state and evaluator identity. If a provider changes an unpinnable model behind an endpoint, comparisons spanning that change are treated as a new observational epoch unless a paired same-time control can recover comparability.

3 Terminology: what "lattice growth" means in Orion

The project name Orion contains "Knowledge Lattice," but v3 does not assert that all persistent research state forms one global mathematical lattice with a single meet/join operation. The canonical substrate is a typed compatibility and relational structure containing epistemic projections, evidence, operators, episodes, obstructions, strategies and meta-method variants. Lattice/concept-lattice views may be constructed locally where the required order/closure structure is defined and useful.

Accordingly, this paper uses *Orion state growth* or *retained structured growth* for the quantitative seven-axis process in Section 11. Informal phrases such as "the lattice grows" are shorthand for growth of this typed persistent research state, not a claim that one global lattice cardinality increases monotonically.

This distinction matters empirically. A new obstruction, a new operator and a new epistemic claim are different typed additions and should not be added into one raw scalar. The seven-axis vector preserves these distinctions, while raw node/edge counts remain inventory measurements.

A local lattice view can have its own domain-specific size, closure or concept-count statistics, but those statistics are reported with the exact local view and cannot be promoted to a global Orion complexity measure.

Measurement coverage is equally explicit. The v3 runtime owns episodes, lessons, tools, failures, saturation and optional evolution state. Legacy epistemic `KnowledgeFiber` projections are included in a metrology snapshot only when the caller supplies the exact fibres that define the measurement universe. The resulting unified-substrate hash covers those supplied objects, while the v3 state fingerprint covers only the v3-owned value state. Neither quantity attests arbitrary repository files or external services.

3.1 Terminology at a glance

For readers from machine learning or systems engineering, the following plain-language glosses fix the load-bearing vocabulary; formal definitions appear at first technical use.

`TaskEpisode` / `Lesson`
: an immutable typed record of one research attempt in one context; and a versioned, *defeasible* abstraction over one or more episodes.

Defeasible
: revisable in light of later evidence; a lesson is a working generalization, not a proof.

Non-sovereign
: a proposal or method change cannot promote itself; canonical/method authority is minted only by a separate governed step.

Fibre
: the research thread/index scoped to a fixed problem and context.

`DifferenceWitness`
: the typed object recording *why* a transfer or reuse was rejected — the specific structural difference that broke it.

meta-QoI
: the quantity a method change predicts it will improve (e.g. a predicted drop in the repeated-failure rate R_F); a Class-B challenger must state one before outcomes are seen.

ORACLE arm
: an upper-bound evaluation arm given oracle assistance, used to test whether the base model can perform the task *at all* before any learning claim is made.

`RESET` vs `LEARNING` arms
: fresh-model-without-learned-method vs with-method arms of the attribution design.

Sham memory
: a budget-matched, structurally-irrelevant memory control that isolates memory *content* lift from mere extra context.

Retained (semantic) novelty
: an addition judged genuinely new after de-duplication against prior state, as opposed to raw growth.

Epistemic non-interference
: the invariant that experience-side state cannot be confused with, or leak into, scientific authority.

`SEARCH/JUMP/GLUE/LIFT`
the escalating retrieval/composition routes: reuse same-domain, transport cross-domain, compose partial results, specify a missing operator.

Status token	Plain meaning
`MODEL_CAPABILITY_FLOOR_`m	at model size m the base model cannot clear the task threshold, so learning effects are unidentifiable there
`CAPABLE_MODEL_AVAILABLE=NO_REFUTED`	no model that clears the gate is yet available; the causal study stays unauthorized
`NEGATIVE_NO_TRANSFER_SUCCESS_LIFT`	a scoped null: no measured transfer-success improvement, not evidence of harm
`DEMOTED_AI_OPERATOR`	an evidence path deliberately denied independent/confirmatory authority
`PROPOSAL_SHADOW_STORED` vs `CANONICAL_INVENTORY_ADMITTED`	recorded as a non-authoritative proposal vs admitted to the canonical record

Table 1: Legend for the governance status tokens used in the results and status tables.

4 Pilot observations from the live research programme

This section reports early qualitative observations from the existing `RAKL_math` research record. The cases were not generated under one uniform v3 telemetry protocol, many were discovered before Paper 5 was defined, and they are not statistically independent. We therefore use them to identify hypotheses and failure classes, not to estimate frequencies or claim causal improvement.

4.1 Retrieval coverage can fail even when memory is correct

The cross-Millennium XM003 audit exposed a particularly informative failure [28]. An earlier synthesis, XM002, stated that Hodge remained an uncalibrated future target for a reusable cross-domain audit operation. XM003 inspected the exact frozen base of that synthesis and found that the Hodge memory review already selected the tool, recorded a target-specific DifferenceWitness, and linked subsequent Hodge calibration/failure artifacts. The evidence existed in the audited state; it was simply not retrieved by the synthesis.

This distinction changes the proposed repair. More memory content would not have helped. The missing object is a coverage certificate that binds the search universe: repository revision, registered lanes/indexes, query coordinates, included and deferred regions, result identities and a re-verification trigger. The application failure therefore produced a framework child issue, Orion issue 119, asking for a bounded cross-problem coverage receipt rather than another retrieval prompt [29].

The lesson is broader than this instance. A local memory review can be internally consistent while a narrative completeness statement is false because the search universe itself was incomplete. A future Orion method should distinguish

`NO_RELEVANT_MATCH_IN_BOUND_UNIVERSE`

from an unbounded prose statement that “no relevant memory exists.”

4.2 Rich surrogate state can be free

P-versus-NP XM004 tested a source-native closure representation proposed for a cover/fusion route [30, 23]. In the all-rectangle source regime of Cavalar and Oliveira, a singleton generator above an edge has empty trace on the complement; upward closure then forces the entire powerset into the local closure. Raw closure cardinality is therefore maximal even though the source’s exact cover-complexity coordinate is zero for that regime.

The mathematical consequence is narrow. It rejects raw closure volume, entropy, or any hardness score that becomes large merely because the generator basis plus upward-closure convention produces many sets. It does not reject the underlying closure representation or more selective pair-indexed productive ancestry.

The research-method consequence is more general. Across hard problems, agents repeatedly construct quantities that are locally meaningful and mathematically clean but are not known to preserve the coordinate required by the root claim. The cheapest adversarial question is often not "can we optimize this quantity?" but

> can this surrogate look maximally good while the root coordinate is zero or wrong?

This motivates a reusable `ROOT_COORDINATE_PRESERVATION_AUDIT` candidate operator for later prospective evaluation.

4.3 The same faithfulness pattern appears in unrelated mathematics

The Riemann-Hypothesis analytic lane reached a structurally similar boundary through Li's criterion. Suzuki constructs concrete L^2 objects whose norms are non-negative, but the exact all-index identity transporting that norm to the Li coefficients is itself RH-equivalent. The resulting atom therefore treats the defect between positive surrogate and target coefficient as the object to audit, rather than treating positivity of the norm as progress toward RH [31, 24].

The RH spectral lane exposed a second representation defect. Under an exact compact-resolvent zero-counting hypothesis, the manuscript context initially suggested ordinary weak-L^1 endpoint behaviour for a first-order resolvent-type operator. An adversarial endpoint calculation found an additional logarithmic divergence and blocked candidate generation before a `CANDIDATE_PROPOSED` event. The frozen packet was preserved and the correction was recorded retrospectively rather than backfilled into the preregistered history [32].

These cases support two different process hypotheses. First, positive or regularized representations need an independently audited faithfulness map to the root quantity. Second, context packets themselves must be falsifiable before candidate generation; a strict chronology is useful only if a defective context can be quarantined without erasing it.

4.4 BSD shows why scalar normalization can hide structural extra zeros

The BSD Selmer/Iwasawa comparison programme asked whether an independently defined discrete or augmentation order can be compared non-circularly with the complex Taylor order at $s = 1$. A source audit of Mazur–Tate/Kato interfaces identified explicit split-multiplicative corrections and examples of extra augmentation zeros, including a good-prime mechanism that can create augmentation vanishing even when Mordell–Weil rank is zero [33, 25].

The route was therefore not repaired by subtracting the desired rank or complex order, which would be circular. Instead the residual asks for independently defined corrections, a no-extra-zero condition, or a richer leading-graded/derived object. For the method case study, this is another instance of representation repair preceding theorem invention.

4.5 A correct local lemma may leave the root essentially untouched

A Yang–Mills spectral candidate provides a clean positive example of local progress [34]. For a positive self-adjoint contraction T, if a dense source set shares one common asymptotic nth-root moment bound $q < 1$, a spectral-projector contradiction implies $\sigma(T) \subseteq [0, q]$. A planted restricted-source example continues to hide a slower mode because the source set is not dense, while a dense source family recovers the true slowest rate.

The result closes an abstract hidden-spectrum logical sub-bridge. It does not establish that the controlled Euclidean Yang–Mills observables map to a dense/cyclic subset of the same

Osterwalder–Schrader reconstructed excited Hilbert space, nor that the rate survives support growth, coupling evolution, lattice-spacing limits or continuum spectral identification. Calling the local lemma "progress toward the mass gap" without carrying these interfaces would overstate what was learned.

This case motivates explicit gluing telemetry. A research architecture should be able to report that a local section is supported while the global section remains unavailable because named interface keys are unverified.

4.6 Equation type and source scope can invalidate familiar downstream tools

A recent Navier–Stokes Type-II source route produces an ancient Euler limit in one branch because the rescaled viscosity tends to zero [35, 26]. That observation blocks an otherwise tempting reuse of parabolic Navier–Stokes backward uniqueness as the closing rigidity theorem. The remaining residual concerns no-incoming-energy or far-field tightness inherited by the Euler limit together with an Euler-specific rigidity argument.

This is not evidence that backward uniqueness is a bad method. It is evidence that a method valid in one chart cannot be transported after a load-bearing equation/interface coordinate changes. In Orion terms, the failure is a scoped transfer obstruction, not a blacklist of the operator.

4.7 Failing closed on missing source text is a useful research action

The constructive Yang–Mills lane identified Bałaban's dedicated averaging-operations paper as the relevant primary source for an actual weak-coupling block, but the accessible material did not expose enough mathematical body to instantiate the nonlinear averaging map, block/path conventions, covariance equations or regularity hypotheses [36, 27]. The lane left the corresponding contract fields unknown and blocked candidate generation rather than reconstructing a generic block map from memory.

In a conventional benchmark this episode might be scored as no progress. In an open research programme it is a useful narrowing of the evidence boundary. Paper 5 therefore counts source-bound blocking and precise residual creation as legitimate research outcomes while keeping them distinct from theorem progress.

4.8 The v3 merge itself is a governance case study

The framework repository provides a direct meta-level incident. Orion was merged in commit `a151d561...` with an explicit message that the merge occurred before CI cleanup at operator direction [37]. A later repair PR fixed post-merge CI problems, including the known floating-point assertion in the v3 experience benchmark and unrelated publication/checkout interactions [38].

This incident is useful precisely because it violates the stronger promotion discipline that Paper 5 advocates. A direct user instruction can legitimately override an automated gate operationally; indeed coding-agent instructions are subordinate to direct operator instructions in many harnesses. But an operational override should not retroactively become evidence that the framework change was validated.

We therefore introduce a distinction between *operational deployment state* and *method authority state*. An override can put code on `main` while leaving the method variant provisional for assurance purposes. Post-deployment failures and repairs become immutable episodes. A later promotion-grade claim must bind the exact active state and fresh evidence rather than infer validity from repository location.

4.9 What the pilot does and does not show

The pilot supports a taxonomy of recurring research-process failures that cuts across mathematical domains:

1. retrieval/search-universe coverage;
2. surrogate-to-root faithfulness;
3. representation and correction terms;
4. local-to-global gluing;
5. method-transfer scope and interface changes;
6. source completeness and provenance;
7. chronology/evaluator/CI identity;
8. operational governance versus epistemic promotion.

It does not establish their prevalence, causal effect on success, or uniqueness to Orion. It also does not show that v3 prevents them, because most episodes were generated before standardized v3 telemetry. Those stronger questions define the prospective evaluation programme in Section 11.11.

5 From raw experience to bounded research method

The purpose of the v3 experience substrate is not to maximize how much past text can be recalled. It is to preserve an auditable relation between what happened, what the system later believes it learned, and what that belief is allowed to change.

5.1 Fast recording and slow consolidation

Every consequential attempt enters the fast loop first:

$$\text{action} \rightarrow \text{observation} \rightarrow \texttt{TaskEpisode}.$$

The episode is frozen before root-cause interpretation. This ordering prevents a later explanation from rewriting the evidence it is supposed to explain.

The slow loop operates on already-recorded episodes:

$$\text{episodes} \rightarrow \text{contrastive analysis} \rightarrow \text{lesson proposal} \rightarrow \text{replay/transfer validation} \rightarrow \text{scoped method update}.$$

A failed fresh transfer bounds or contradicts a lesson rather than being averaged away. Promotion creates a new lesson version; it does not mutate the source episodes or silently replace the old lesson.

This architecture responds to a known failure mode of agentic memory. Continually rewriting consolidated memory can degrade performance even when the underlying experiences were useful, while retaining raw episodes can remain competitive [9]. Orion therefore treats consolidation as a defeasible derived view rather than the canonical memory root.

5.2 Authority of a lesson

A lesson can occupy a bounded authority ladder such as

$$\texttt{CANDIDATE} < \texttt{VERIFIED\_LOCAL} < \texttt{CONDITIONALLY\_REUSABLE} < \texttt{PROOF\_BACKED},$$

with `SUPERSEDED` representing historical lineage rather than a higher state. The order is not automatic. A candidate reflection cannot become reusable because the same LLM says it sounds general. A reusable lesson requires independently registered verification and a fresh successful transfer under the relevant implementation contract, or a separately valid proof-backed route.

The current v3 implementation still contains authority-bearing surfaces whose caller-supplied identifiers and flags require stronger content-bound resolution before promotion-grade use. Paper 5 therefore treats new v3 experience in the live mathematical programme as proposal/shadow evidence until these paths are hardened.

5.3 Failure learning is an evidence ladder

Failure learning uses a deliberately slower promotion chain:

$$\texttt{TaskEpisode}(\text{non-success}) \to \text{observed failure} \to \text{competing diagnoses} \to \text{discriminating test} \to \text{supported diagnosis} \to \text{boundary lesson candidate} \to \text{cross-context validation.}$$

The reason is causal. A failed proof attempt may have failed because the theorem is false, because a representation was weak, because the source was incomplete, because a tool was unavailable, because the context packet was wrong, or because CI failed. Treating the first plausible narrative as the causal obstruction would poison future routing.

Diagnosis revisions are append-only versions. A later diagnosis can supersede an earlier one without erasing the original observation. A raw failure cannot directly mint a globally blocking impossibility; only an independently verified impossibility can block reuse, and only inside its registered scope.

5.4 Experience affects routing, not theorem truth

For operator o in target context c, an experience statistic can summarize matched episodes, successes, partial successes, failures and costs. One illustrative routing score is

$$\text{score}(o \mid c) = C(o) + D_V(o) + R_B(o) - \lambda_s \widehat{p}_s(o \mid c) + \lambda_f \widehat{p}_f(o \mid c) - \lambda_e U_o,$$

where C is execution cost, D_V verification debt, R_B boundary risk, $\widehat{p}_s$ and $\widehat{p}_f$ smoothed success and failure/block rates, and U_o a small exploration bonus for undersampled operators. The exact coefficients are a policy choice to be evaluated, not epistemic constants.

The hard rule is structural:

$$\boxed{\text{experience-conditioned priority} \neq \text{epistemic authority}}.$$

A high empirical success rate cannot weaken a proof obligation. A historically poor operator may still be used if the current DifferenceWitness restores a previously broken assumption. This avoids turning experience into a self-reinforcing blacklist.

5.5 Strategy motifs and expertise

Repeated successful contiguous operator sequences can be proposed as strategy motifs. For example, a cross-domain motif may take the form

$$\text{freeze root coordinate} \to \text{identify surrogate} \to \text{construct hostile zero-root control} \to \text{audit preservation map} \to \text{rotate representation if the map fails.}$$

The motif is not accepted because it occurred frequently. Contradictory episodes containing the same sequence remain attached to the motif, and applicability is context-indexed.

This separates two kinds of reusable knowledge that recent systems often combine under "skills." Agent Workflow Memory learns reusable routines [5]; AutoSkills constructs multi-level skill hierarchies [6]; and XSkill separates action-level experience from task-level skills [7]. Orion uses a similar intuition but adds evidence lineage, falsifiers, authority boundaries and cross-context contradiction as first-class fields because the target application is open-ended research rather than only task completion.

5.6 Problem fibres as retrieval state

The problem fibre couples the experience loop to the active research state. For each atom it co-retrieves relevant knowledge, tools, failures, episodes, motifs and expertise. This makes later method analysis possible at a finer level than "memory on/off." A researcher can ask:

- Was a relevant prior failure present in the bound universe?
- Was it retrieved?
- Was it rejected, and on what structural DifferenceWitness?
- Did the chosen operator already have a failure boundary in the target context?
- Did the fibre omit a whole problem lane because the coverage index was stale?
- Did the compiled context exceed the budget and drop a mandatory atom?

These questions convert memory performance into inspectable process variables rather than one opaque retrieval score.

5.7 Learning from gluing failure

Suppose a problem is decomposed into atoms $A_1, \ldots, A_n$ and local candidate sections s_i are verified. A global solution requires more than

$$\forall i \ \text{Verified}(s_i).$$

The sections must cover all required atoms and agree on shared interfaces. Let I_{ij} denote the set of declared shared interface coordinates between A_i and A_j. Then a simplified compatibility condition is

$$\forall x \in I_{ij}, \qquad s_i(x) = s_j(x),$$

plus dependency and authority conditions.

In research, the interface may be a uniform constant, a limit theorem, a category-preserving realization map, a source-density result, or a statement that a local invariant actually controls the root quantity. A gluing obstruction therefore has a typed residual signature rather than being summarized as "proof incomplete."

The pilot cases suggest that gluing failures may be unusually common in hard open problems. Yang–Mills exhibits fixed-cutoff versus continuum and source-to-spectrum interfaces; BSD exhibits discrete/p-adic versus complex analytic coordinates; Hodge exhibits cohomological detection versus algebraic realization; Navier–Stokes exhibits compactness/limit-class versus rigidity hypotheses. Paper 5 preregisters gluing-failure detection as a process QoI because earlier recognition may save large amounts of unproductive local theorem search.

5.8 Saturation and the invention gate

The system should not infer "missing operator" merely because repeated attempts fail. Invention is proposed only after a bounded saturation audit supports all of the following:

1. relevant knowledge/operator/path axes are flat under the registered search window;
2. the residual is stable across repeated independent routes;
3. ordinary causes such as missing source detail, wrong context, unavailable tool, implementation failure or verification debt have been excluded;
4. cross-domain transfer routes have been searched or bounded;
5. explicit evidence supports a representation or method-basis gap.

The resulting report permits escalation to an invention process. It does not prove that invention is necessary. This fail-closed distinction is important for studying "scientific creativity." A system that invents new formal objects whenever it is confused may look creative while simply failing to retrieve or compose what already exists.

5.9 Orion-triviality as an empirical hypothesis

Let $N_R(P)$ denote the minimum genuinely novel problem-solving structure required for a verified solution of problem P relative to Orion state R. If existing resources, operators and motifs suffice through composition or transfer, then $N_R(P) = 0$. For a solved distribution D we can define

$$\rho_R(D) = \frac{|\{P \in D : N_R(P) = 0\}|}{|D|}.$$

This is not assumed to be high. It is a measurable hypothesis about the relationship between the growth of reusable method space and the growth of problem space.

The longitudinal observatory extends the question from terminal solutions to research progress. A problem may remain unsolved while a useful subproblem is classified as transfer-novel or representation-novel. Over many episodes, the distribution of novelty classes can reveal whether the system's bottleneck is missing knowledge, missing representations, missing operators or merely poor routing among already-known components.

5.10 Unified substrate and Self-Orion

The v3 unified substrate materializes a read-only typed overlay over specialized stores. Epistemic items, operators, episodes, failures, strategy motifs and architecture variants can be related without transferring semantic authority to the overlay itself. This lets Self-Orion represent its own variants as meta-method objects while preserving the original stores that own proof, failure or promotion semantics.

The intended recursion is therefore not

$$\text{LLM rewrites itself and declares success.}$$

It is

$$\text{observed research} \rightarrow \text{bounded method hypothesis} \rightarrow \text{challenger variant} \rightarrow \text{matched evaluation} \rightarrow \text{fresh assurance} \rightarrow \text{governed promotion or rejection.}$$

The next sections make this engineering loop explicit.

6 Obstruction–transformation episodes as a consolidation plane

The semantic-shortcut layer adds a structural projection over the experience substrate. Its unit is a source episode

$$O \xrightarrow{T} O',$$

where the obstruction O is represented through load-bearing roles, relations, constraints, failure mechanisms, invariants, desired transitions and forbidden losses, while T records a transformation and its source preconditions. The resulting record retains provenance, observed effects, preserved invariants, broken constraints, breakpoints, source authority and content identity.

This object is deliberately different from both a raw `TaskEpisode` and a promoted research tool. A raw episode says what happened in one execution context. An obstruction–transformation episode is a structural retrieval view over such evidence. A `ResearchTool` additionally asserts a scoped reusable method contract. These three objects may share lineage but do not share authority automatically:

$$\text{observed episode} \neq \text{structural transformation episode} \neq \text{promoted reusable tool.}$$

6.1 Search before invention

For a target obstruction, the router searches the frozen episode memory in the order

$$\texttt{SEARCH} \to \texttt{JUMP} \to \texttt{GLUE} \to \texttt{LIFT},$$

with `CANNOT_CHECK` when evidence for the next transition is insufficient. Same-domain complete transformations are preferred; cross-domain reuse requires an explicit target mapping witness; partial verified effects remain composition candidates; and an admissible composition must bind interfaces, order and incompatibility checks.

This order turns experience memory into an anti-invention pressure. Repeated failure by itself does not justify another primitive. The system must first ask whether a relevant transformation is already present, whether a distant structural analogue survives transfer, and whether partial known transformations can be composed safely.

6.2 Repeated residuals and inverse invention

If SEARCH, JUMP and GLUE are boundedly exhausted, the system still does not infer that a novel operator is necessary. LIFT additionally requires the retrieved candidates to be accounted for, a cross-problem coverage receipt for the no-match claim, and at least two distinct failed attempts sharing a residual feature. The output is then an inverse design target:

$$\mathcal{S}_{\text{missing}} = (\text{preserve}, \text{break}, \text{expose}, \text{reduce}, \text{validate}).$$

It specifies what an invented representation/operator would have to accomplish without asserting that such an object exists or that its eventual implementation will work.

This strengthens the basis-gap discipline developed earlier in the paper. Saturation is no longer only a qualitative reason to consider invention; for strict mathematical discovery it is tied to a content-identified search universe, explicit candidate rejection and repeated residual structure.

6.3 Post-success consolidation

A successful target candidate does not automatically write itself back as universal experience. Consolidation happens only after target validation at the authority actually achieved. If the target transformation is worth preserving, Orion may add a new structural episode with exact source/target lineage. If the method step is independently reusable under a scoped contract, it may separately become or update a `ResearchTool`. Contradictory and failed target attempts remain attached as negative history.

The long-horizon loop is therefore

$$\text{record} \to \text{retrieve/map/compose} \to \text{residual} \to \text{bounded specification when needed} \to \text{validate} \to \text{scoped consolidation.}$$

This loop can alter future proposal order and available abstractions while leaving theorem/scientific authority under the ordinary verification path.

6.4 Promotion-grade routing evidence and its boundary

The routing plane is no longer only a deployed proposal architecture. A versioned RSHEA campaign separated three questions that the earlier matched-model experiment had bundled: whether the typed route state contains the distinctions needed for safe routing; whether those distinctions are enforced by production code; and whether an LLM can infer/use the state well enough to improve research outcomes. The first two are now measured, while the third remains a separate capability-limited causal coordinate.

A 21-family, 2,688-case adversarial validation of the strict typed router was previously quoted for this plane. A repository audit (`research/paper3_gate_falsifiability_audit_v1/UNREPRODUCIBLE_V2_RESULT.json`) found that although its protocol freeze survives, no harness, case set or result artifact for it exists anywhere in the repository or its history; the quoted numbers are therefore unreproducible and carry `CANNOT_CHECK` status—not a refutation—and this paper does not cite them as evidence. The audit also records a structural prediction, made before any such harness exists, that the instrument shares the self-grading gold-assignment defect confirmed in the sibling experience-to-action harness (its four registered thresholds sit exactly at the ceilings its metrics could not leave). The failed first validation instrument for this plane is separately preserved: it was rejected because two intended mutations were not load-bearing. The routing conformance evidence this paper does rely on is the exact-head production test surface described below.

The same semantics have been carried into a stacked production implementation. The first resolver preserves `SEARCH`→`JUMP`→`GLUE`→`LIFT` priority and delegates every positive route to the canonical audit. A second successor introduces content-bound rejection certificates for conclusive `SEARCH/JUMP` failures so an unresolved earlier candidate cannot be bypassed. A third introduces exact-composition rejection certificates for `GLUE`: episode order and revisions, target/context, memory snapshot, composition-witness content, canonical audit reasons and evidence pointers are bound, component episodes are not deleted, and `LIFT` is attempted only after every current GLUE composition is either accepted or individually certified rejected. Missing, ambiguous or non-conclusive composition evidence remains `CANNOT_CHECK`. Exact-head routing tests, the protected repository test workflow and artifact-contract coverage are green for the final stacked candidate.

These results license a scoped method-governance claim, not a generic continual-learning or model-performance claim. The earlier model-dependent matched evaluation remains preserved as `CANNOT_EXECUTE_MATCHED_EVAL` at the registered capability boundary; it is not reinterpreted as a routing win. Thus the paper can claim that the obstruction–transformation memory has a promotion-grade, fail-closed routing and negative-history mechanism, while whether model-mediated use of that memory improves fresh scientific research remains causally unidentified at the available operating point.

7 Unified solver mechanics as method-evolution objects

The unified problem-solving projection adds several proposal-only method objects—operational-map coverage, path/concurrency quotienting, multiobjective path cost, fieldability, Mechanic Differential Diagnosis (MDD), Verified Solver Compilation (VSC) and trajectory-to-certificate assembly. They do not constitute a second experience system. Their executed decisions, costs, residuals and verification outcomes are recorded through the existing `TaskEpisode` substrate and may become `Lesson` or ResearchTool candidates only through the consolidation and fresh-assurance process defined in this paper.

This matters most for MDD and VSC. A diagnosis that labels a stalled route as a representation, metric, scale, verifier or operator-basis problem is merely a hypothesis until a discriminator or downstream repair provides evidence. Likewise, a compiled representation–solver–decoder–verifier configuration is a method candidate, not a capability upgrade. Repeated development success may justify a scoped procedural lesson such as "this transformation effect reduces search in this structural regime," but promotion requires held-out transfer, cost accounting, negative-history preservation and the ordinary protected method-evolution gate.

The learning target should therefore remain vector-valued. Useful telemetry includes cause-identification accuracy, specialist conditional advantage, representation/portal preservation, geometry construction and reuse cost, closed-loop routing success, verifier scheduling cost, staleness and root-level residual reduction. A single end-to-end win cannot retroactively attribute every component. Conversely, a failed geometry or solver compilation becomes structured negative history that may reveal a missing operator, ontology coordinate or context boundary. This makes the unified solver a consumer and producer of method-evolution evidence without making it a sovereign evaluator of its own improvement.

8 Implementation status at the Paper 5 evidence cutoff

Because this paper combines deployed v3 machinery, integrated measurement instrumentation and prospective evaluation, Table 4 states the implementation boundary explicitly. The source-level matrix is maintained in `docs/PAPER5_IMPLEMENTATION_STATUS.md`. "Deployed" or "implemented" never means that a fresh-task capability improvement has been demonstrated.

Table 4: Status of major Paper 5 surfaces at the current evidence cutoff. Implementation, deployment, internal hardening, empirical improvement and independent assurance are distinct coordinates.

Surface	Status	Licensed statement
TaskEpisode/Lesson substrate, failure revisions, problem fibres, saturation, novelty classes, unified substrate	deployed implemented	v3 can persist, relate and query typed research experience; persistence alone does not establish fresh-task improvement
Episode storage admission	deployed, internally hardened	`PROPOSAL_SHADOW_STORED` retention is distinct from `CANONICAL_INVENTORY_ADMITTED`; shadow history cannot satisfy a protected canonical consumer by filename or path convention

Surface	Status	Licensed statement
Experience-conditioned routing and strategy motifs	deployed implemented	experience can alter search ordering; benefit and transfer validity remain empirical coordinates
Protected lesson/tool promotion	deployed, internally hardened	authority-bearing procedural promotion resolves protected content-bound evidence rather than relying only on caller narratives; independent external assurance is not claimed
Certificate-backed local-section/gluing authority	deployed, internally hardened	local verification and global gluing authority are subject-bound and fail closed under the implemented protected path; mathematical correctness still depends on the registered evidence/checker boundary
Prospective pre-action chronology	deployed, internally hardened	consequential learning-turn execution can require a content-bound pre-action fibre/operator/falsifier receipt; absent valid prospective binding is reported as retrospective rather than receiving prospective-evidence credit
Cross-problem memory coverage	deployed,	
bounded-universe contract	cross-problem "no relevant match" claims require a bound coverage receipt; this does not license unbounded retrieval	
Epistemic noninterference specification	composed executable invariant	current `RAKLV3State` carries an immutable `ScientificAuthorityProjection`; registered scientific promotion, revocation and supersession own the authority coordinate while experience/retrieval/routing/workspace/self-evolution operations are tested for noninterference. Historical pre-#242 `NO_INTEGRATION_SURFACE` is preserved as superseded negative/incomplete history; this is not a model-level low-leakage or superiority result
Optional training-time projection	proposal-only architecture; no scheduler efficacy	$\pi_{\text{train}}(R_t, \theta_t)$ can bind checkpoint/probe-specific structural mastery and derived training candidates while remaining distinct from search rank and scientific authority; no structural-saturation or training-efficiency claim is licensed until #461/#466 execute

Surface	Status	Licensed statement
Evolution archive and governed promotion binding	deployed, internally hardened	variants can branch while preserving rollback targets; governance/evolution claims are bound to protected subjects rather than a bare Boolean on the current protected path
Matched RESET/LEARNING experience benchmark	executed scoped negative	v1.2 native job 3476548 produced 12/12 schema-valid records with correct chronology but zero registered successes in both arms under Qwen2.5-0.5B-Instruct; ORACLE floors continue through 0.5B (3476730/3476731), 1.5B (3476756; instrument defect 3476742 preserved), 3B (3476778), 7B (3476788; success 1/3, parse-valid 3/3) and V2_0_EXEC sealed revisit 3476813 (success 2/5, parse-valid 5/5; `MODEL_CAPABILITY_FLOOR_7B_V2_EXEC`); `CAPABLE_MODEL_AVAILABLE=NO_REFUTED`; no transfer-success lift or global capability claim is licensed
Paper 5 state/process metrology	integrated measurement-only	read-only quantitative snapshots, seven-axis retained-growth accounting, typed process telemetry and cost-policy identity can be recorded; metrology has no promotion authority
Four-arm causal-attribution objects and scripts	integrated, preregistered; confirmatory freeze/execution terminal refusal	#250 `CANNOT_FREEZE_CONFIRMATORY_PACKET` after shared #247 capability floor; #251 `CANNOT_EXECUTE_FROZEN_PACKET`; executor+harness are instrument-only; no four-arm causal performance result is reported
Method-state manifest and upgrade protocol	integrated governance map	coding agents can identify the incumbent method epoch, classify changes and construct challengers; the manifest itself grants neither promotion nor evolution evidence
Independent retained-novelty audit	protocol frozen, evidence pending	internal seven-axis novelty accounting is measurable but is not represented as externally validated semantic novelty
Independent external assurance of v3 hardening	pending	current implementation closes known declaration-bound authority paths under internal tests; independent/process-lineage-separated assurance remains a stronger unperformed claim
Orion 3.1 superiority or incumbent status	not established	a future Class-B challenger must earn that label through matched development, fresh protected assurance and governance

This separation is itself part of the method. A new commit can change deployment or implementation status, but it cannot automatically convert a prospective empirical claim into an evidence-supported capability statement. The current two-arm negative result is therefore reported as a real constraint rather than hidden by subsequent framework changes, while the four-arm confirmatory study is terminal-refused and the independent novelty audit remains human-blocked on #255. The optional training projection similarly changes what can be represented and measured, not what training-time efficacy has been demonstrated.

9 A governed upgrade constitution for recursive research systems

The central engineering risk in self-improvement is semantic collapse between four distinct events:

$$\text{code exists} \neq \text{code is deployed} \neq \text{improvement is supported} \neq \text{the method is the governed incumbent.}$$

A software branch can be implemented and even merged for operational reasons while its scientific method claim remains unvalidated. The live v3 deployment supplies an instructive example: the v3 branch was merged into `main` under an explicit operator request before its pending CI had completed, and later commits repaired package/publication CI. That sequence is preserved as process history. It is not represented as evidence that v3 was superior merely because it was deployed.

9.1 Method identity is not the package version

The inspected Python package still reports software version `0.1.0`. We therefore separate software/API compatibility from research-method identity. A complete experimental citation should bind at least

$$V = (v_{\text{pkg}}, v_{\text{method}}, e_{\text{constitution}}, \Sigma_{\text{schemas}}, h_{\text{git}}, h_{\text{validators}}).$$

We use `3.0.0` as the method identity for the deployed v3 architecture at the Paper 5 freeze, while the exact Git subject remains the decisive reproducibility identity. Patch-level method versions such as 3.0.x are reserved for implementation corrections that do not intentionally change research policy. Minor method versions such as 3.1 and 3.2 denote workflow/method challengers. A change to protected authority/governance semantics is constitutional and should advance a separately visible constitution epoch or major method generation rather than being hidden in an ordinary workflow patch.

The Paper 5 branch introduces a proposal-only `RAKL_VERSION.json` manifest to make these coordinates machine visible. Its presence does not promote itself to canonical governance.

9.2 Change classes

We retain the existing three-way classification in `meta.py`.

Class A: implementation. Examples include parser defects, CI portability, serialization, hashing, deterministic replay and publication build repairs. A Class-A change requires a reproducing regression, relevant invariant tests, artifact identity and preservation of history. It can improve engineering reliability without supporting a method-capability claim.

Class B: workflow or method. Examples include retrieval policy, fibre compilation, routing, saturation, gluing, lesson induction, memory consolidation and research-portfolio allocation. A Class-B challenger must state a positive meta-QoI prediction before evaluated outcomes, run matched parent/challenger development tests, include fresh transfer and hostile near-misses, preserve blocking invariants, use comparable resources, and pass protected fresh assurance before a strong evolution claim.

Class C: constitution. Examples include changing what evidence may mint authority, weakening review independence, changing the meaning of missing evidence, or changing who can promote an incumbent. Constitutional changes are never auto-promoted. They require explicit human-visible amendment review, migration/rollback analysis and adversarial authority tests.

9.3 Roles and anti-self-certification

A strong evolution claim separates at least seven roles, even if one organization operates them:

1. **observer** records content-bound research episodes and process telemetry;
2. **upgrade proposer** maps repeated evidence to a falsifiable framework hypothesis;
3. **challenger engineer** implements the frozen proposal;
4. **development evaluator** runs visible debugging/replay tests;
5. **fresh assurance evaluator** uses a packet frozen before mutation and hidden from the optimizer;
6. **governance authority** decides whether an assured variant becomes incumbent;
7. **post-promotion attestor** observes the actual active repository state and exact-main validation.

Same-session role labels do not create independence. Process independence and evidence-lineage independence must be separately established when independent credit is claimed.

Proposition 1 (No self-promotion from proposal output)**.** *Suppose a challenger cannot modify the protected evaluator, assurance packet, promotion thresholds or governance credentials that judge it, and promotion requires a content-bound certificate issued outside the challenger path. Then the challenger's generated narrative or code output alone is insufficient to establish its own promotion.*

Proof. The promotion predicate contains at least one required input whose value is controlled outside the challenger and is bound to the exact challenger identity. Changing the candidate output cannot set that external input to an accepting value. Therefore self-generated output is insufficient for promotion, even if it can affect development metrics. The claim is conditional on the protection and identity assumptions; compromise of those boundaries remains a separate threat model. □

9.4 Upgrade proposal packet

Before implementation outcomes are observed, a Class-B/C proposal freezes a machine-readable packet containing the parent method/Git identity, source episode and diagnosis IDs, change class, alternative diagnoses, proposed mechanism, affected method contracts, predicted primary and secondary meta-QoIs, material-effect thresholds, possible regressions, negative controls, hostile near-misses, development benchmark, fresh assurance reserve, model/tool/resource contract and rollback plan. If the result predates the packet, it is retrospective calibration only.

This packet is intentionally stricter than an issue description. It converts the statement "we think this might help" into a falsifiable prediction about a content-identified framework mutation.

9.5 Challenger lifecycle and variant DAG

We model evolution as a DAG rather than a destructive sequence:

$$\begin{gathered}\texttt{OBSERVED\_PATHOLOGY} \to \texttt{UPGRADE\_HYPOTHESIS} \to \texttt{PREREGISTERED} \to \texttt{CHALLENGER\_IMPLEMENTED} \\ \to \texttt{DEVELOPMENT\_VALIDATED} \to \texttt{FRESH\_ASSURANCE\_PENDING} \to \{\texttt{ASSURED}, \texttt{REJECTED}, \texttt{META\_OVERFIT}, \texttt{CANNOT\_CHECK}\} \\ \to \texttt{GOVERNANCE\_APPROVED} \to \texttt{PROMOTED} \to \texttt{ACTIVE\_PROMOTION\_ATTESTED} \to \texttt{MONITORED}.\end{gathered}$$

Rejected and superseded variants remain negative history. A previous incumbent remains a rollback target when compatible.

The existing v3 `EvolutionArchive` already separates challenger, assured, rejected and incumbent states and does not auto-promote a challenger merely because an evolution trial passes. However, its current `promote_incumbent` interface accepts a caller-provided `governance_approved` Boolean. We regard that as an implementation gap, not as adequate promotion evidence. A future 3.1 challenger should replace declaration-bound governance with a content-bound attestation naming exact candidate, assurance verdict, constitution epoch and approving process.

9.6 Development evaluation versus fresh assurance

Two evidence phases answer different questions. Visible development evaluation can guide debugging and optimization; once exposed, it is no longer blind assurance. Strong method-evolution evidence therefore consumes a separate fresh packet frozen before the challenger and hidden from the proposer. Repeated peeking consumes the assurance exposure budget. The existing `SelfEvolutionAssessor` and `evaluate_bootstrap_trial` already distinguish development gain, fresh transfer, evidence-lineage independence, evaluator separation, resource matching and meta-overfit. These components support a scoped verdict rather than a global "Orion evolved" state.

9.7 Evaluator migration

A reviewer may reasonably object that fixed evaluators eventually become obsolete. We allow evaluator evolution, but not inside the candidate it judges. A legitimate evaluator change is a separately governed parent mutation. Within an evaluation epoch, the ruler is frozen. At epoch boundaries, a new evaluator can be proposed, benchmarked against known-answer and adversarial cases, versioned and then used prospectively. This differs from allowing the optimizer to rewrite its reward after seeing an inconvenient score.

9.8 Operator override and deployment before assurance

Human or repository governance can intentionally override the scientific promotion process. The protocol therefore models an explicit `OPERATOR_OVERRIDE` with requested deviation, exact subject, blockers known at the time and containment/rollback plan. An override may move code if repository permissions allow, but creates no evolution-evidence credit. The deployed variant remains method-provisional until the required evidence is produced. This rule is not hypothetical; it is motivated by the actual v3 merge-before-CI incident and prevents the paper from retrospectively rewriting operational history into a clean preregistered story.

9.9 Post-promotion attestation and rollback

Pre-promotion authorization is not evidence that deployment occurred correctly. The existing `promotion_attestation.py` distinguishes a candidate that passed checks from an active state whose `main` ref descends from the candidate, required content matches and exact-active-main post-promotion validation succeeds. A promoted variant also carries an exact rollback parent, migration notes, rollback test and monitoring conditions. New evidence can narrow, supersede or roll back a variant without erasing the evidence that originally supported it.

9.10 Version increments as empirical claims

Under this constitution, a label such as `3.1` has scientific content. It means that a distinct workflow/method challenger has been identified and evaluated under the registered process; it does not mean that a certain number of commits were merged. Consequently Paper 5 can plot

ORION 3.0, 3.1 and later versions as experimental interventions rather than arbitrary software milestones.

10 Informing coding agents: repository-native method state and Codex harness

Recursive framework governance is ineffective if the coding agent modifying the package does not know which method version is incumbent, which rules are protected, or how a proposed change will be evaluated. Conversely, putting every historical fact into a giant system prompt consumes context and quickly becomes stale. We therefore use the repository itself as the durable system of record and make the coding-agent entrypoint a navigation layer.

10.1 Instruction hierarchy

OpenAI's Codex documentation specifies that repository `AGENTS.md` files can provide coding conventions, repository structure and testing instructions; their scope follows the directory tree, more deeply nested files can specialize broader instructions, and direct system/developer/user instructions take precedence [20]. Codex's published harness-engineering guidance further argues for giving the agent a map rather than a very large instruction manual, with deeper repository documentation serving as the source of truth [21]. We adopt that hierarchy rather than assuming the agent carries a persistent private memory of the latest ORION architecture.

The Paper 5 challenger adds a compact self-evolution entrypoint to the existing root `AGENTS.md`. For a task that can change framework behavior it points to:

1. `RAKL_VERSION.json`, a machine-readable method-state manifest;
2. `docs/RAKL_UPGRADE_PROTOCOL.md`, the governed challenger lifecycle;
3. `docs/RAKL_METROLOGY.md`, the frozen measurement vocabulary;
4. `src/rakl/method_specs.py`, the canonical process contracts;
5. protected promotion/evolution/attestation surfaces relevant to the requested change.

The intent is not to make `AGENTS.md` the constitution itself. A future engineering cleanup should further reduce the root file toward a table-of-contents role while preserving exact deeper sources of truth.

10.2 Startup protocol for a framework-editing Codex session

Before editing a framework method, a coding agent should execute the following observable startup sequence:

1. resolve the current exact `main` Git SHA and method-state manifest;
2. classify the request as Class A, B or C;
3. locate the affected method contract and protected evaluator/authority surfaces;
4. inspect the relevant case-study episodes, diagnoses or lessons motivating the change;
5. freeze the upgrade proposal, target metrics, benchmark/evaluator identities, negative controls and rollback plan when Class B/C;
6. create or use a challenger branch rather than silently rewriting the incumbent;

7. implement the smallest change that tests the stated mechanism;
8. run exact-subject tests and emit a machine-readable handoff containing parent, candidate, metrics, blockers and rollback identity.

This sequence is intentionally inspectable from repository artifacts. We do not require disclosure of hidden model chain-of-thought.

10.3 What happens when a user directly instructs Codex to bypass the protocol?

The instruction hierarchy means a direct user/operator request can override repository guidance [20]. The method therefore cannot rely on `AGENTS.md` as an access-control mechanism. The correct response is architectural rather than rhetorical. The repository records an operational override and withholds evolution-evidence credit. Protected CI, branch protection, external assurance and post-promotion attestation remain separate controls where configured. Thus the scientific statement is not "Codex cannot disobey the method"; it is "an override is observable and does not become evidence of method improvement merely because the code moved."

10.4 Method-state manifest

The proposal-only `RAKL_VERSION.json` introduced on the Paper 5 branch names the current method generation, exact Git subject at proposal freeze, software package version, constitution epoch, protected evaluation surfaces and known upgrade gaps. A canonical future form should also bind active schema versions, evaluator bundle hashes, migration status and last attested promotion.

The manifest solves a practical coordination problem. A newly started Codex session, human developer or other coding agent can determine the intended incumbent without relying on conversational memory. At the same time, the exact Git SHA prevents a friendly version label such as "3.1" from hiding divergent code.

10.5 Package/API version versus method version

Software consumers need ordinary API compatibility semantics, whereas scientific evaluation needs method identity. These are different concerns. A serialization bug might require a Python package patch without changing research behavior. Conversely, a routing-policy change might constitute a new research method even if public Python APIs remain backward compatible. The manifest therefore records both. Release artifacts should additionally use the repository's content-addressed release manifest so that binaries or generated documents can be tied to an exact source subject.

10.6 State and schema migration

Persistent experience means framework upgrades cannot treat state migration as an implementation footnote. A new version must state whether old episodes, lessons, failures, tools and saturation records are:

- byte-compatible and directly readable;
- translated by a content-bound deterministic migration;
- retained only as legacy read-only evidence;
- or incompatible and therefore excluded from the challenger evaluation.

A migration must never manufacture higher authority. If a new schema can express stronger authority than the old one, imported objects retain the old effective authority until independently revalidated. Migration tests include round-trip identity where promised, source-pin preservation, supersession lineage and rollback compatibility.

10.7 Concurrent agents and branch isolation

Long-running research may involve several solution agents, a meta-observer and one or more coding agents. Concurrent work creates race conditions in both software and epistemic state. The safe default is immutable source episodes plus isolated challenger branches. A framework engineer should rebase or merge current `main` before final evaluation, recompute exact subject hashes and rerun candidate checks. A benchmark packet bound to an earlier head cannot be silently claimed for a later integrated head.

The same principle applies to mathematical applications. `RAKL_math` remains the application-state repository; reusable framework mutations belong in `Orion`. Application evidence may motivate a framework issue but cannot write itself directly into framework authority.

10.8 Reproducibility under model-service drift

Repository state can be exactly versioned; hosted model behavior may not be perfectly reproducible even under the same public model name. Strong experiments therefore record model identity/revision when available, prompt hashes, temperature, seed where supported, token ceilings, tool policy, external retrieval policy and wall-time/resources. If an exact model revision is unavailable later, the replication is a new condition rather than an exact rerun. Framework evolution claims should be robust enough to survive matched tests across more than one model/revision when the claim is intended to generalize beyond a specific model.

10.9 Why the coding agent is part of the experiment

The coding agent is not merely infrastructure. Its engineering behavior supplies additional Self-Orion case evidence: evaluator drift, CI repair, migration defects, stale instructions, over-large context files, unsafe authority interfaces and rollback friction are all observable framework-process outcomes. Paper 5 therefore treats engineering incidents as `TaskEpisode`-like evidence at the meta-method level rather than cleaning them from the historical record once fixed.

11 Quantifying growth, process quality and causal assistance

A recursive research architecture is not supported by the observation that its repository grows or that its underlying language model eventually succeeds. A system can accumulate duplicated or irrelevant memory, spend more compute, and still attribute the model's unaided ability to the framework. We therefore treat metrology as part of the method rather than as post hoc visualization. The measurement contract is implemented prospectively in `docs/RAKL_METROLOGY.md` and the read-only `src/rakl/v3_metrology.py` surface on the Paper 5 challenger branch. These measurements grant no scientific or promotion authority.

11.1 Seven-axis retained-growth vector

The state-growth coordinates reuse the v3 saturation axes. For research cycle t we define

$$g_t = (\Delta K_t, \Delta O_t, \Delta E_t, \Delta B_t, \Delta R_t, \Delta P_t, \Delta M_t),$$

where the components denote retained novelty in knowledge, operators, experience patterns, obstructions, relations, successful paths and meta-methods, respectively. The qualifier *retained* is essential. Raw files, commits, nodes or generated lessons are inventory, not evidence of learning. The v3 `NoveltyRound` contract records retained semantic novelty and separates it from repeated route searches or paper count.

Alongside g_t, each frozen state reports node counts by substrate kind, edge counts by typed relation, episode outcomes, lesson and tool authority distributions, failure-diagnosis status, evolution-variant status and unresolved cross-view links. For states S_t and S_{t+1}, `compare_state_metrics` produces a content-independent growth delta. We do not collapse these quantities into a single "lattice size" score.

Several ratios diagnose whether growth is useful rather than merely large. We preregister lineage coverage as the fraction of derived canonical objects with valid source ancestry, unresolved-link rate in the common substrate, supersession-preservation rate, and a semantic novelty-retention ratio that compares retained additions with raw proposals once the necessary proposal identity stream is available. High raw growth with low lineage coverage or high orphan rate is a negative result, not progress.

11.2 Experience conversion and failure learning

The fast v3 loop stores immutable `TaskEpisode` records; the slow loop may propose lessons, diagnoses, tools and strategy motifs. We measure the conversion funnel rather than assuming that every stored episode is useful:

$$\text{episodes} \rightarrow \text{supported diagnoses or lessons} \rightarrow \text{reusable tools or motifs} \rightarrow \text{validated fresh reuse.}$$

Reported quantities include episode outcome rates and costs, lesson proposal and validation yields, tool reuse success and failure rates, failure-diagnosis maturation, time from observation to supported diagnosis, obstruction-resolution rate, and the repeated-failure rate

$$R_F = \frac{\#\{\text{failures with a previously observed structural signature}\}}{\#\{\text{failures}\}}.$$

A useful experience system should reduce R_F on fresh tasks without increasing false transfer. Neither a high lesson yield nor a low lesson yield is intrinsically desirable. Excessive abstraction can be as harmful as failure to learn.

Table 5 lists the subset of these quantities an implementer can instrument on their own long-running agent *today*, independent of the (unrun) causal-attribution study. They are process-health measurements, reported as a profile, not a single "method score".

11.3 Process telemetry for the full method surface

Orion already declares canonical process surfaces in `method_specs.py`. Paper 5 requires consequential invocations to be measurable. A proposal-only `ProcessTelemetry` record binds the method surface, input and output state identities, active task and fibre, outcome, registered cost, residual transformation, retrieved/selected/rejected object identities, verification artifacts and the retained novelty vector. The common aggregate for process p reports

$$N_p, \quad P_p(\text{success}), \quad P_p(\text{blocked}), \quad P_p(\texttt{CANNOT\_CHECK}), \quad \overline{C}_p, \quad \overline{\Delta|B|}_p,$$

plus axis-specific retained novelty and object-selection counts. Here B denotes the registered residual set; raw residual-count contraction is descriptive because replacing one vague blocker by several precise obligations can represent epistemic progress despite increasing $|B|$.

The process-specific dashboard covers decomposition, routing, search-query generation, source qualification, claim extraction, ontology normalization, context translation, equivalence/structural transfer, contextual gluing, contradiction diagnosis, gap discovery, discriminator selection, synthesis, memory, review, benchmarking, authority promotion, saturation stopping, context compilation, capability shaping, software execution, research-portfolio allocation, objective evolution and generator transport. The complete definitions are frozen in the metrology contract. Representative quantities include atom reopen rate and late interface discovery for decomposition; saturated-route retry rate for routing; retrieval precision and recall in a bound universe for memory; hostile-near-miss rejection and false-transfer rate for transfer; local-success/global-failure rate for gluing; pre-promotion defect capture for review; exact-subject replay for software execution; and invalid promotion rate, whose target is zero.

11.4 Residual contraction as a common progress coordinate

Open research often lacks a scalar objective before the target theorem or artifact exists. We therefore record progress through typed residual transformations. Let B_t be the canonical active blocker set for an atom. A simple contraction

$$C_t = |B_t| - |B_{t+1}|$$

is reported only when blocker identity is stable. The more informative representation records blockers resolved, newly exposed, unchanged and reopened. This prevents the framework from manufacturing "progress" by renaming an obstruction or hiding a root-like obligation inside a new representation.

11.5 Four-arm causal-attribution benchmark

The key empirical question is whether ORION contributes anything beyond the base model. We therefore extend the earlier reset-versus-learning design with a separate four-arm attribution benchmark (Figure 2). The same frozen task packet, model revision and registered resource ceiling are used in every arm:

`MODEL_ONLY` The same base LLM and permitted external tools operate without the ORION workflow or persistent ORION state.

`RAKL_RESET` The ORION workflow is active, but every evaluation task starts from the same initial framework state. This estimates the static architecture effect.

`RAKL_SHAM_MEMORY` The workflow and memory/context budget are matched, but relevant learned memory is replaced with registered irrelevant or structurally incompatible controls. This tests whether gains arise from experience content rather than merely extra context or preprocessing.

`RAKL_LEARNING` The full workflow uses the learned state produced by the development sequence. Every fresh-transfer task starts independently from the same frozen learned state.

For registered metric m we report separate lift vectors

$$\Delta_{\text{architecture}}(m) = m(\texttt{RAKL\_RESET}) - m(\texttt{MODEL\_ONLY}),$$

$$\Delta_{\text{experience}}(m) = m(\texttt{RAKL\_LEARNING}) - m(\texttt{RAKL\_RESET}),$$

$$\Delta_{\text{content}}(m) = m(\texttt{RAKL\_LEARNING}) - m(\texttt{RAKL\_SHAM\_MEMORY}),$$

and

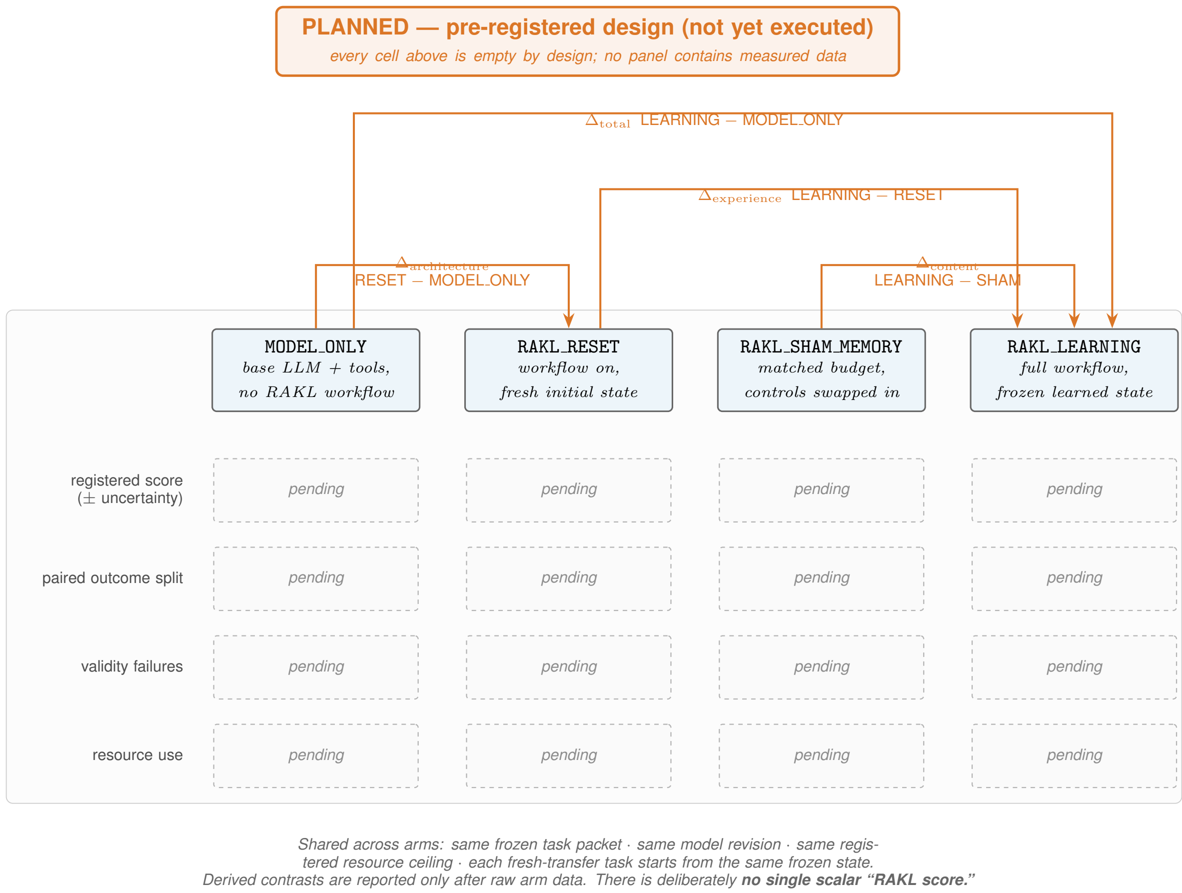

Figure 2: The preregistered four-arm causal-attribution design (*planned; not yet executed*). The arms `MODEL_ONLY`, `RAKL_RESET` (static architecture, fresh state), `RAKL_SHAM_MEMORY` (budget-matched irrelevant memory) and `RAKL_LEARNING` isolate four separate lift contrasts: architecture (RESET – MODEL_ONLY), content (LEARNING – SHAM), experience (LEARNING – RESET) and total (LEARNING – MODEL_ONLY). Result cells are empty by design: this is a registered design, not a result, and there is deliberately no single scalar "Orion score."

$$\Delta_{\text{total}}(m) = m(\texttt{RAKL\_LEARNING}) - m(\texttt{MODEL\_ONLY}).$$

There is deliberately no scalar "Orion score".

For binary task success, each matched pair is also assigned to `BOTH_SUCCESS`, `RAKL_ONLY_SUCCESS`, `BASELINE_ONLY_SUCCESS` or `BOTH_FAIL`. The asymmetric counts are scientifically important. A `RAKL_ONLY_SUCCESS` is direct scoped evidence of assistance; a `BASELINE_ONLY_SUCCESS` is direct evidence that the framework interfered and must be reported with equal prominence.

11.6 Efficiency and epistemic-safety lift

Framework contribution is not limited to final success. When both arms solve a task we compare model tokens, preprocessing tokens, retrieval calls, tool calls, wall time, candidates attempted, failed route families and time to the first decisive falsifier. A framework that spends substantially more resources for the same result is not credited as a capability gain unless the claim is explicitly about a quantified quality or safety tradeoff.

For open research, epistemic-safety effects may appear earlier than root solutions. We therefore measure unsupported scope escalation, false theorem/candidate promotion, invalid structural transfer, surrogate-to-root coordinate errors, false global gluing, chronology violations and source-scope errors. A framework that lowers false progress while holding solution rate fixed may still be scientifically useful, but that claim is kept separate from capability improvement.

11.7 Decision-level contribution witnesses and component ablation

Population-level matched trials establish the main attribution. To understand mechanism, a consequential intervention may emit a proposal-only contribution witness recording what Orion consulted, how route ranking changed and what action was selected. Examples include failure memory demoting a repeated route, a retrieved tool promoting a cross-domain method, a root-coordinate preservation gate blocking a seductive surrogate, or a gluing check exposing a missing interface.

Such a witness is not causal proof. Where feasible, we preregister component ablations or leave-one-memory-out replay. Candidate ablations include removing failure memory, success-tool memory, experience-conditioned routing, problem-fibre compilation, gluing checks, saturation/invention gates, or a specific high-impact lesson. Repeated seeds or sampling configurations are required before interpreting stochastic replay differences as stable effects.

11.8 Statistical units, dependence and uncertainty

Tokens and tool calls are not independent observations. The primary unit is the task for matched outcome comparisons, the episode for process/failure measurements, the problem family or domain for transfer generalization, and the framework version for evolution claims. Development episodes are temporally dependent by design because the state changes. Fresh-transfer tasks must all begin from the same frozen learned state so that transfer task T_1 cannot train T_2.

We use paired analyses whenever the same task can be evaluated under multiple arms. Binary success is reported through paired contingency counts and an appropriate paired interval/test. Continuous scores and resource quantities are analysed as paired differences with bootstrap, permutation or model-based uncertainty appropriate to the frozen sampling design. If sufficient task families accumulate, hierarchical analyses will separate within-family from cross-family effects. Primary and secondary metrics, material-effect thresholds, multiplicity policy, resource ceilings, stopping rules and assurance exposure budgets are frozen before evaluated results.

11.9 Goodhart resistance and non-compensatory invariants

Once a metric influences upgrades it becomes a target. We therefore prohibit one-number promotion. Benchmark and evaluator hashes, thresholds and hard authority invariants are protected from the challenger they judge. Raw storage growth never counts as improvement, hostile near-misses expose over-transfer, and a gain in success or efficiency cannot compensate for an authority, history, evaluator-integrity or fresh-transfer leakage failure. If an upgrade legitimately changes a metric or evaluator definition, that evaluator migration is treated as a separately governed parent change rather than being allowed to redefine the ruler inside the challenger.

11.10 What would demonstrate that Orion is doing real work?

We preregister a minimum evidentiary chain for the statement that the framework learned or assisted:

$$\boxed{\text{retained semantic growth} + \text{traceable downstream reuse/decision effect} + \text{matched baseline lift} + \text{fresh transfer when generalization is claimed} + \text{no compensating validity regression}}$$

Repository size, issue count, prose volume, proposed lesson count, or an agent's self-report that ORION was helpful do not satisfy this chain. The strongest version-level claim additionally requires protected fresh assurance and governed promotion under the upgrade protocol in Section 9.

11.11 Historical model precursor, root-cause correction, and the active experience mechanic

The earlier matched experience benchmark remains a useful negative precursor for this paper's model-adapter programme. After two preserved response-interface failures, the frozen v1.2 RESET-versus-LEARNING packet completed native job 3476548 with twelve of twelve schema-valid records and the intended chronology: RESET tasks begin from the registered initial state; LEARNING development forms a single $S_0 \to S_n$ chain; and every fresh-transfer LEARNING task starts independently from the same frozen S_n.

On the staged Qwen2.5-0.5B-Instruct model, both arms achieved zero registered successes on both development and fresh-transfer tasks. The development score delta is zero, and the small partial-score difference on fresh transfer does not satisfy the registered success criterion. The bound historical result therefore remains exactly `NEGATIVE_NO_TRANSFER_SUCCESS_LIFT_UNDER_0_5B`. The earlier v1 and v1.1 executions likewise remain in the lineage as prompt-interface failures rather than being pooled with the interpretable v1.2 result. Nothing in the later RSHEA loop relabels these records as positive.

The subsequent preregistered ORACLE ladder also remains immutable history: 0.5B, 1.5B and 3B produced 0/3 registered successes; 7B produced 1/3; and the sealed V2 revisit at 7B produced 2/5 with five of five parse-valid outputs. Under the then-active packet those runs did not clear the 2/3 operating-point rule. Figure 3 records that historical ladder exactly.

RSHEA did not stop there. A later Stage-0/1 audit of the evidence-binding instrument found a construct defect: the prompt/evaluator contract had not defined the semantic roles of the selected- and rejected-evidence-ID fields, and two apparent model failures were exact inversions of those undefined roles. This diagnosis does not invalidate the fact that the historical packet failed its frozen gate; it invalidates using that packet as permanent authority that model reasoning itself is the required bottleneck. A later hosted GLM-5.2 probe on the repaired conceptual surface reached 30/30 exact conceptual passes in both arms, clearing the conceptual 2/3 capability floor, while simultaneously showing that the old single-task panel had become a ceiling and could not discriminate RAKL from the control. The remaining model-adapter residual is therefore fresh evidence relevance/binding under a discriminating panel, not a justification for making `CAPABLE_MODEL_AVAILABLE=NO_REFUTED` a permanent dependency of the method architecture.

The active method mechanic is tested at the structured boundary instead. Once an experience object has been verified, scoped and content-bound, the research question is whether the typed experience state contains information needed to select a legitimate fresh action while refusing stale, refuted, out-of-scope or insufficient cases.

What the first structured instrument measured, and what it could not. A frozen 1,792-case / 14-family minimal-twin benchmark compared typed selective experience against RESET, failure-memory-only, scalar-confidence, provenance-only, untyped whole-state and their strongest composite projection. A subsequent governance audit of that instrument (`research/paper3_gate_falsifiability_audit_v1/`) found that its harness assigned the gold action and the candidate prediction from the same pure function on the same argument, so four of its six registered gate conditions—exact action accuracy, unsafe-apply rate, `CANNOT_CHECK` recall and legitimate-apply recall—were satisfied *by identity* and could not fail under any perturbation of the evidence (a black-box probe battery left all four unmoved in 32/32 trials per probe while the two live conditions moved under the identical probes). Those four numbers are therefore not candidate measurements and are not reported as such here. The original receipt is preserved verbatim as negative instrument history. What the benchmark validly measured is an *information-sufficiency and projection-lossiness* result: the registered typed coordinates determine the governed action by construction, while the five named simpler projections provably do not—the strongest composite simple parent reaches a deterministic information ceiling of 0.714 with unsafe-apply rate 0.286 and zero `CANNOT_CHECK` recall, quantities that are real, falsifiable properties of those projections. All ten registered hostile mutations were detected.

The repaired instrument. The successor gate severs gold from the candidate. An independently implemented declarative oracle (`experiments/paper3/independent_action_oracle_v1.py`; statically audited to neither import nor reference the candidate) assigns gold after case freeze on a fresh 2,400-case stratified panel whose records expose only the fourteen candidate-visible coordinates; the dataset's construction labels are forbidden oracle inputs. Under this instrument the candidate's exact action accuracy, unsafe-apply rate, `CANNOT_CHECK` recall and legitimate-apply recall become measurable quantities, and each of the four is demonstrated falsifiable by at least one planted candidate mutant. A per-condition black-box falsifiability audit of the repaired gate (`REPAIRED_GATE_FALSIFIABILITY_AUDIT.json`) returns FALSIFIABLE on all eight evidence-dependent gate conditions, with controls asserted first: the recorded receipt reproduces exactly; a deliberately re-planted self-grading variant reproduces the historical NON-FALSIFIABLE signature under the identical probe set; and a doctored oracle fails the independence audit. The observed conformance result—accuracy 1.000, unsafe-apply rate 0, `CANNOT_CHECK` recall 1.000, legitimate-apply recall 1.000, composite-parent information ceiling 0.433, 12/12 mutants caught—is an *implementation-conformance* statement about the shipped structured-state mechanic against an independently derived specification. Because the oracle was authored in the same repository process, this is redundant derivation, not independent review; the claim it supports is conformance plus the information-sufficiency separation, never model capability.

This successor chain changes the active dependency graph rather than rewriting the model negative. The required ORION experience-to-action transition is the typed and selective structured-state mechanic, now held to a gate that has been shown capable of failing. Natural-language lesson extraction and model-level fresh-task lift are adapter/empirical-assurance questions: they may improve or fail under future frozen panels, but they do not mint method-state or scientific authority and they cannot disable an already verified structured-state transition. The fresh model/interface qualification lane therefore remains scientifically useful, but it is no longer a failed required runtime node.

Status of the fresh-task lift. The registered four-arm fresh-task benefit obligation remains `BLOCKED_ON_CAPABILITY_QUALIFICATION`: the frozen Stage-4 protocol requires an A100-class subject executing the exact pinned unquantized 7B loading semantics, and whether to stage that subject is an operator budget decision. This is a prospective, unexecuted measurement—explicitly not a null and not a negative—and no threshold of the frozen protocol has been relaxed to route around it. An externally constructed staleness benchmark (STALE, arXiv:2605.06527) was identified as the strongest lever for an external non-self-generated evaluation surface; a feasibility verification (`research/paper3_stale_feasibility_v1/`) found its published full-context protocol infeasible under the currently sanctioned local envelope (every scenario requires a ∼167–200K-token context and hosted LLM judging), while its incremental memory-system mode remains a viable, separately frozen prospective lane.

The result still does not substitute for a completed four-arm model-level treatment study. It does not establish universal LLM continual-learning efficacy, CONTENT-specific sham-memory lift, or superiority of one hosted model. Its purpose is narrower and load-bearing: preserve the old failures, identify why they were not a valid stopping rule for ORION, and replace the

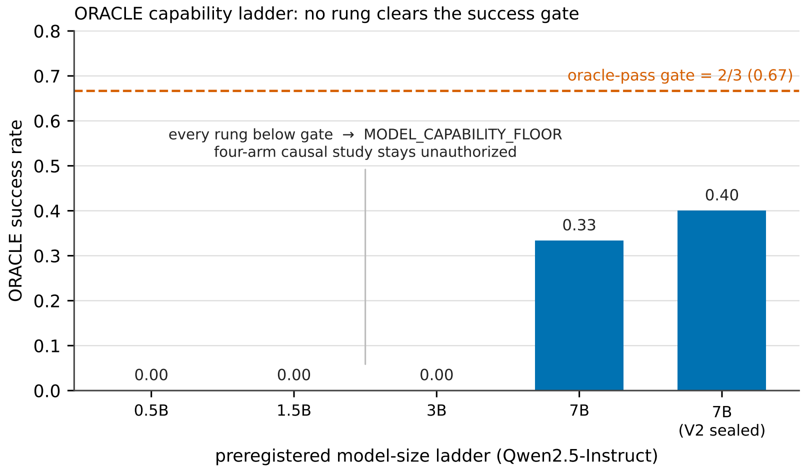

Figure 3: Historical preregistered ORACLE ladder for the ExperienceBenchmark upper-bound arm. Under the then-frozen small-panel gate, none of the 0.5B–7B runs reached the 2/3 pass threshold. These points remain immutable negative/capability-floor history. Subsequent RSHEA audit identified an evidence-role construct defect in the old interface and later hosted capability evidence cleared the conceptual floor while ceilinging the old task; accordingly this figure is no longer used as authority that model capability is a required dependency of the structured method-state mechanic.

required dependency with a mechanic that earned promotion under its own frozen adversarial gate.

11.12 Measured cautionary evidence from the structural-learning lane

The programme's structural-learning lane (its Paper IV artifacts, which remain intact and byte-unchanged in the research record) reached a terminal its own routing gate classified as conceptual cross-paper evidence rather than a standalone result. We absorb it here in that capacity, because both of its findings are direct measurements about the risks of *method evolution itself* — the subject of this paper. Nothing in this subsection promotes a structural-learning law; the defensible objects are two negatives and one instrument mechanic.

An adaptive method-evolution candidate lost to its static parent, with attributed cause. A learner-state-conditioned adaptive training allocator — exactly the kind of candidate a method-evolution loop generates — was compared against a static structural parent at strictly equal example budget across six frozen simulation worlds (384 replicates). The adaptive arm lost on balanced mastery by -0.0166 (bootstrap 95% CI $[-0.0174, -0.0158]$) and degraded the hard-safety floor. Failure attribution then localized the loss to the allocation-policy stage via two designed lever arms: *guard-rail budget capture* (a coverage floor implemented as a budget-consuming target rather than a constraint absorbed 13 of 48 examples into the highest-mastery coordinate in every world; capping it recovered about a third of the gap) and *within-round concentration* (near-uniform budget still lost -0.0108). One diagnostic arm was found confounded and is recorded as uninformative rather than used as evidence. The originally recorded root-cause narrative of the parent receipt was refuted by the realized budget counts and is superseded *as an interpretation only*; the receipt itself is immutable and preserved. Both identified defects were verified present in the promoted production allocator, so the diagnosis transfers beyond the simulator. The operational consequence stands: the static parent retains default authority, and a preserved negative about our own adaptive mechanics is part of this paper's method-evolution evidence, not an embarrassment to be routed around.

The instrument itself could not have adjudicated the question — provably. The deeper finding is about the evaluation instrument. A three-tier ceiling analysis showed that *no* equal-budget allocation policy, however optimal, could have reached the instrument's own registered material-effect gate of 0.05: a greedy-oracle policy achieved +0.0015 (a lower bound), local search found +0.0045 (a lower bound), and a rigorous harm-free-relaxation upper bound caps every policy at ≈0.0246 — a factor of two below the gate, stated with its tightness caveat because an earlier, stronger version of this bound was itself found invalid and corrected before use. Conventional power analysis was silent throughout: bootstrap intervals were ∼0.0016 wide, amply "powered" for an effect the instrument could not generate. An independent lane that tuned allocation policies on the same instrument class plateaued at the constructive ceiling computed here and stopped — corroborative, cross-instrument agreement between a mechanism bound and an empirical search exhaustion.

What this changes for governed method evolution. These are two instances of one lesson, and it is the same lesson taught by the gate-falsifiability audit of Section 11.11: before a confirmatory budget is spent, the evidence instrument must itself be qualified. Falsifiability audit asks whether a gate *can fail*; admissibility analysis asks whether the instrument *can express the effect it is gating.* The structural-learning lane contributes the second mechanic in executable form: compute the instrument's oracle ceiling before an allocation comparison and require it to clear a pre-frozen multiple of the minimum detectable effect, emitting `INSTRUMENT_INADMISSIBLE_CEILING_BELOW_GATE` — and spending nothing — otherwise, failing closed to `CANNOT_CHECK` when the oracle is not computable. This paper's governed upgrade protocol adopts that requirement prospectively for its own evaluation epochs: a method-evolution challenger may not consume a confirmatory epoch on an instrument whose ceiling has not been shown to clear the registered effect gate. The fresh-task lift protocol of this paper is explicitly flagged by the same logic: its 0.05 gate has no recorded ceiling qualification, and executing it without one risks buying an underpowered terminal with scarce accelerator hours.

12 Security, governance and failure containment

A self-modifying research framework is also an adversarial system. The relevant adversary need not be a malicious external actor. An optimizer that legitimately seeks a higher score can accidentally exploit stale evaluators, contaminated memory, ambiguous identities or permissive authority interfaces. We therefore treat the model, retrieved text and challenger code as untrusted proposal sources with respect to protected authority.

12.1 Trusted computing base and its boundary

The intended trusted computing base is deliberately smaller than the research agent. It includes the protected Constitution/authority rules, exact-subject identity checks, evaluator and assurance packet identities, promotion/attestation logic, and the repository or execution mechanisms used to observe these facts. The language model, same-session reviewer roles, generated lessons, retrieved documents and challenger implementation are not trusted to certify themselves.

Content hashes establish identity under the assumed hash function; they do not establish semantic truth. A Git commit proves neither that the mathematical claim is correct nor that the evaluator is well designed. Likewise, a signed governance attestation would establish who approved an exact object, not that the approval was scientifically wise.

12.2 Memory poisoning and prompt injection

Persistent experience creates an attack surface absent from stateless agents. A malicious or simply erroneous episode can be repeatedly retrieved and gain influence through consolidation.

Retrieved webpages, papers, issues and code comments may contain imperative language that attempts to act as instructions. Orion therefore separates evidence content from agent-control instructions. External content can support claims or propose actions, but cannot alter the Constitution or coding-agent instruction hierarchy merely by being retrieved.

Derived lessons preserve supporting and contradicting episode lineage. A lesson is not allowed to delete the raw evidence from which it was distilled. This design is motivated both by general provenance principles and by recent empirical work showing that repeated LLM memory consolidation can degrade useful memories even when the underlying episodes remain informative. A later contradicted or superseded lesson remains inspectable rather than overwriting the episodes that expose the error.

Hostile memory tests should include fabricated high-confidence lessons, duplicated episodes, contradictory success/failure histories, stale-context reuse, malicious instruction text embedded in evidence, and a lesson whose apparent support consists of many copies from one lineage.

12.3 Evidence Sybils and pseudo-independence

Counting artifacts creates an obvious Sybil vulnerability. Ten summaries of one source do not constitute ten independent sources, and ten role labels in one context do not constitute ten independent reviews. Metrics therefore count source/evidence lineage and process independence rather than raw citation or agent count. Repeated failures derived from one copied episode cannot manufacture a cross-context obstruction pattern; repeated successes from the same lineage cannot manufacture transfer breadth.

12.4 Evaluator hacking and Goodhart pressure

An evolving agent will eventually discover regularities in any visible benchmark. The protection strategy is layered:

- development tasks are explicitly reusable and may be optimized against;
- strong assurance tasks are frozen before the challenger and hidden from the proposer;
- assurance exposure is budgeted, and exposure converts a packet into development evidence;
- evaluator code, thresholds and subject-identity rules are protected from the challenger they judge;
- hostile near-miss tasks measure over-generalization;
- no scalar metric can compensate for an authority or integrity failure.

A legitimate evaluator evolution is possible only through a separate governed change between evaluation epochs. This matters because recent self-improving-agent research has begun to explore co-evolving evaluators. Our design does not assert that a fixed evaluator should remain optimal indefinitely; it asserts only that the candidate must not rewrite its own ruler within the trial used to claim improvement.

12.5 Authority forgery and protected subject binding

The first v3 integration audit identified a concrete defect class: caller-provided booleans and opaque string identifiers could be mistaken for high-authority evidence. The current implementation has since hardened the known lesson/tool, local-section/gluing, evolution/governance and benchmark authority paths with protected content-bound subject hashes, attestation resolution and fail-closed bindings. Thus the historical defect is retained as negative engineering evidence rather than presented as a property of the current code.

The governing invariant is

$$\text{authority transition} \Rightarrow \text{resolved exact subject} + \text{licensed evidence type} + \text{protected gate} + \text{scope compatibility}.$$

Internal conformance and hostile tests can show that the implemented path enforces this contract for registered attacks. They do not prove that the trust roots themselves are uncompromised, that every future call site is correct, or that an external evaluator would reproduce the assurance claim. Residual risks therefore include stale or misbound registry entries, protected-key compromise, evaluator contamination, operator override and ordinary implementation defects. Paper 5 distinguishes implementation hardening from independent external assurance.

12.6 Chronology attacks and backfilling

If an agent first observes a result and then creates a "preregistered" packet that predicts it, the evaluation is contaminated even when every file is internally consistent. Hash-chained traces and frozen packet identities make this harder to hide, but only if the upstream chronology is actually observed. Retrospective results are useful calibration and learning evidence; they simply receive zero prospective credit.

The live case study contains several chronology/provenance repairs, including research results that were downgraded to retrospective calibration rather than being backfilled into a strict discovery trace. We treat this as a positive governance behavior and a process-cost signal at the same time.

12.7 Cross-task leakage

Persistent learning is supposed to transfer development experience to fresh tasks, but transfer tasks must not teach one another inside the same assurance packet. The v3 benchmark therefore freezes the learned state after development and starts every fresh-transfer task independently from that exact state. A run whose initial state already contains a prior transfer result invalidates the packet.

The same principle applies across benchmark arms. The sham-memory condition must be constructed before target outcomes and must not accidentally contain semantically equivalent answers or target-specific hints.

12.8 Supply chain, external tools and model services

Orion depends on Python libraries, Git hosting, CI runners, external search services and hosted models. Exact repository hashes do not pin all of these dependencies. Strong release/evaluation receipts should therefore record environment and dependency identities where material, and distinguish an unavailable dependency from a scientific refutation. A supply-chain compromise or hosted-model behavior change can invalidate reproducibility without falsifying the method hypothesis.

12.9 Denial of service through memory growth

A persistent system can fail by becoming too large to query. Storage growth is separated from prompt growth; problem fibres and context compilation create bounded working sets. Metrology records useful-context fraction, retrieval miss rate and context size. Compaction or pruning may create derived views, but raw canonical episodes remain evidence roots unless a separately governed retention policy applies. Resource stress tests should include memory stores large enough that naive full-context loading is impossible.

12.10 Governance attacks and operator powers

Repository administrators or users with sufficient privileges can bypass software gates. We do not claim cryptographic enforcement against the human owner. Instead the scientific record distinguishes governed promotion from operator override. An override is explicitly logged, cannot self-create evolution evidence and triggers post-deployment containment/attestation work. This boundary is important for honest evaluation of real engineering practice, where emergencies and direct operator instructions exist.

12.11 Rollback as a first-class operation

Rollback is not failure erasure. A promoted variant retains the evidence that supported it, while new incidents can produce a new assessment that narrows or reverses the operational decision. The previous incumbent or another assured branch remains addressable in the variant DAG. Rollback tests bind state/schema compatibility and required active paths. If state migration is irreversible, that fact is itself a Class-B/C deployment risk that must be disclosed before promotion.

12.12 Security success criteria

For Paper 5, the strongest security claims are negative invariants rather than claims of perfect safety. The target measurements include zero invalid promotions in the registered tests, zero false independent-review credit, zero fresh-transfer state leakage, exact-subject evaluator binding, complete preservation of source episodes through lesson supersession, and successful detection of planted authority/identity attacks. A failure on one of these properties blocks a strong framework-evolution claim even if task performance improves.

13 Relation to agent memory, self-improvement and autonomous research

Paper 5 sits at the intersection of several rapidly developing areas. We therefore avoid a broad priority claim such as “the first self-improving research agent.” The contribution is a particular combination of research-process representation, longitudinal metrology, causal attribution and governed architecture evolution.

13.1 Learning from episodic experience

Reflexion showed that language agents can improve without parameter updates by storing verbal feedback in episodic memory and reusing it on later trials [2]. ExpeL similarly extracts natural-language insights from collections of agent experiences and recalls them at inference time [3]. These systems establish the basic usefulness of external experiential memory.

Orion differs in the authority and lineage semantics of its memory. A raw `TaskEpisode` is an immutable evidence root. A `Lesson` is a versioned abstraction with support, contradiction, scope, falsifier and explicit authority. Failure observation, failure diagnosis and reusable obstruction are separate states rather than one reflection string. This separation is motivated by the possibility that a useful episode can support a faulty consolidation. Recent work reports exactly such a failure mode: repeated LLM consolidation can degrade memory utility even when retaining the underlying episodes remains competitive, motivating explicit gates around consolidation rather than mandatory rewriting after every interaction [9].

13.2 Workflows, procedural memory and reusable skills

Voyager demonstrated an ever-growing executable skill library with environment feedback and self-verification in Minecraft [4]. Agent Workflow Memory induces reusable workflows from past trajectories and reports transfer across tasks, websites and domains [5]. More recent systems such as MemSkill make memory-extraction and consolidation routines themselves evolvable, while other skill-centric systems manage creation, reuse and refinement across a skill lifecycle [10, 11].

Our question is complementary. We treat reusable workflows and skills as one view of a broader research state that also contains negative history, unresolved obstructions, compatibility relations, problem-conditioned fibres, gluing failures and meta-method variants. The paper further asks how to measure whether such accumulated structure changes research behavior relative to the same base model, and how a lesson about research practice can safely become a change to the framework that stores and uses future lessons.

13.3 Recursive research loops

AREX alternates deep research with constraint-wise auditing and targeted follow-up research, using a learned context-update tool to maintain a compact improvement state over long horizons [12]. This discovery–verification loop is close in spirit to obstruction-guided research. Orion adds a distinct governance question: the verified state used to improve research is not automatically allowed to certify a modification of the research method itself. Method evolution goes through a separate challenger/evaluator/assurance/promotion path.

13.4 Automated design and recursively self-improving agents

Automated Design of Agentic Systems frames agent design itself as a searchable programming problem and demonstrates meta-agent discovery of transferable agent designs [13]. Gödel Agent makes the agent logic self-referential and editable [14]. Darwin Gödel Machine maintains an open-ended archive of self-modified coding agents and empirically selects improvements on coding benchmarks [15]; Hyperagents extend editable self-improvement to the mechanism that generates future agent changes [16]. These systems make clear that self-modifying code and open-ended variant archives are technically viable research objects.

The Orion variant archive borrows the general lesson that one should preserve multiple branches rather than repeatedly overwrite a single incumbent. Our emphasis, however, is not unconstrained open-ended search. The evidence target is scientific research methodology, where outcome verification can be incomplete, novelty and truth are distinct, and evaluator contamination or authority overclaim is itself a failure. Accordingly, Orion distinguishes development gain, fresh assurance, governance approval and post-deployment attestation, and it preserves operator overrides as non-validating operational events.

A recent Red Queen Gödel Machine explicitly studies co-evolving agents and evaluators under non-stationary utilities [17]. This creates an important comparison. We do not assume that evaluators must remain fixed forever. Instead, we freeze an evaluator within one evidence epoch and require evaluator evolution to occur as a separately governed transition between epochs. This prevents a candidate from changing the criterion that judges the same candidate while still permitting the utility surface to evolve prospectively.

13.5 Evaluator-guided scientific and mathematical discovery

AlphaEvolve combines LLM-generated code with iterative automated evaluation and has been applied to algorithmic and mathematical discovery [18, 19]. This work demonstrates that strong evaluators can turn model generation into productive large-scale search. Paper 4 of the Orion programme addressed the adjacent problem of mathematical authority: proof, formalization, novelty, research value and verifier trust must not be collapsed merely because evaluator-guided search is powerful. Paper 5 moves one level upward and studies the research method as the object being learned.

13.6 Long-horizon behavioural studies of research agents

A particularly close 2026 study follows roughly one hundred sequential architecture experiments performed by a single LLM researcher and finds phase-structured productivity, a long saturation wall, recovery after expanding the action surface, workflow-induced greedy search and rediscovery of established results [1]. That result materially narrows the novelty claim available to this paper: observing a long-running agent and concluding that workflow design affects research behavior is not new by itself.

Our longitudinal case study is intended to add three elements. First, every consequential trajectory is represented in a typed evidence system that distinguishes raw episode, diagnosis, lesson, obstruction, reusable operator and framework variant. Second, the case study is coupled to explicit lattice/process metrology and matched model-only/reset/sham-memory/learning attribution rather than only retrospective behavioural interpretation. Third, recurring process pathologies can trigger content-identified framework challengers whose claimed improvement must pass protected evaluation and governance before becoming a new method version.

13.7 The specific claim of Paper 5

The contribution is therefore not any one ingredient in isolation. The paper studies the closed but non-self-certifying loop

$$\boxed{\text{research episode} \to \text{bounded diagnosis} \to \text{lesson/method hypothesis} \to \text{framework challenger} \to \text{matched evaluation} \to \text{fresh protected assurance} \to \text{governed promotion} \to \text{new research episodes}}$$

while preserving the previous evidence and method variants. Whether this loop produces sustained improvements across several Orion versions is an empirical question. The architecture and early case-study incidents are available now; the stronger longitudinal performance claim remains prospective.

14 Novelty boundary for the semantic-shortcut loop

The obstruction–transformation layer does not justify a broad priority claim. Classical derivational analogy and case-based planning already studied stored problem-solving experience, source–target correspondence, replay/adaptation, failures, repairs and partial-plan reuse; the internal prior-art record names Mostow (1989), Hammond (1990) and Veloso (1994) as representative parents. Modern LLM-agent work likewise establishes experiential learning, reusable workflow/skill memory and iterative feedback-driven refinement [2, 3, 5, 6, 10]. The bounded audit also checks recent structure-aware proof retrieval and counterexample/property-guided synthesis as close modern neighbors. These parent families anticipate substantial parts of the mechanism even when their terminology differs.

Accordingly, the claim available to Orion is narrower. The candidate contribution is the particular executable, fail-closed composition of:

1. content-bound obstruction–transformation episodes with source authority, preconditions, effects, invariants, breakpoints and provenance;
2. vocabulary-light obstruction fingerprints for retrieval;
3. invention-last `SEARCH`→`JUMP`→`GLUE`→`LIFT` routing;
4. explicit target structural/precondition witnesses and forbidden-loss checks;

5. complete-effect semantics for SEARCH/JUMP and partial-effect semantics for GLUE;
6. candidate-accounted, cross-problem-bounded exhaustion plus repeated residual structure before LIFT;
7. a LIFT output that is a missing-transformation specification rather than a self-authorized tool; and
8. continued separation between proposal/search memory and scientific, theorem, novelty or method-promotion authority.

A bounded internal primary-source audit through 12 August 2026 did not identify one source containing this same complete contract. That result is not a global firstness certificate: terminology differs across planning, case-based reasoning, analogy, theorem proving, program synthesis and agent memory, and a combination of strong parent systems anticipates much of the conceptual space. The full source identities, component comparison and strongest-parent attack are frozen in the committed prior-art audit record. Independent external reviewers were unavailable and were not invented; issue #403 closes under `INTERNAL_PRIOR_ART_AUDIT` without minting independent novelty authority.

Paper 5 therefore claims an integrated method/governance design, not invention of analogy, experience memory, repair or compositional reuse in isolation. If a future independent external prior-art review finds one prior system that materially contains the complete contract, the novelty language should narrow further to an implementation/integration contribution without changing the empirical questions in this paper.

Nearest-work occupancy at the 2026-08-14 cutoff. A bounded same-context nearest-work audit with primary-source verification (`research/paper3_gate_falsifiability_audit_v1/NEAREST_WORK_AUDIT.json`; not independent review) found the paper's two claim signatures occupied. The method-evolution signature is contested by verified red-level parents: CoEvoSkills occupies the self-evolving verified-method-library architecture, and Consistency-Gate (arXiv:2607.22962) occupies write-time admission gating of agent memory via LLM self-consistency scores, releasing benchmarks and a gate implementation that any Paper III admission-gating claim must now beat at matched cost. The narrowed information-sufficiency claim is occupied by a classical parent: it is a finite instance of MDP state abstraction. Neither occupancy is a retreat trigger; both are absorbable. The discriminator, and now the single load-bearing novelty axis of this paper, is that ORION's promotion signal is *evidence-bound* — a typed gate over verification, scope, version, refutation, lineage, context, failure signature, strategy family and negative history, returning `CANNOT_CHECK` as a first-class action — where every parent found uses a learned or model-derived score with no refusal semantics. That axis is stated here exactly as the audit leaves it: it is currently untested against any of these parents, and a matched-cost head-to-head against the self-consistency admission baseline is the registered obligation it creates.

15 Threats to validity, alternative explanations and decisive falsifiers

A paper about self-improving research systems is unusually vulnerable to circular evidence. We therefore state the strongest alternative explanations and the outcomes that would force claim narrowing.

15.1 The live case study is observational and selected

The six Millennium programmes were intentionally chosen because they are difficult, open and structurally diverse. They are not a random sample of scientific research. Their value is as a stress-test observatory, not as a population from which we can directly estimate the average behavior of all research agents. Furthermore, individual atomic tasks are selected adaptively by the agents and framework. Frequencies observed in the live archive therefore mix properties of the problem distribution with properties of the selection policy.

Paper 5 accordingly separates retrospective pathology counts from prospective benchmark results. Statements such as "surrogate/root-coordinate confusion recurs across several domains" are case-study observations. Population-level rates require a frozen sampling rule or benchmark set.

15.2 Open problems do not provide frequent final success labels

For an unsolved root problem there may be no positive terminal outcome during the study. This creates a risk that the framework rewards internally generated notions of progress. Residual contraction, lesson authority and local theorem validation are therefore reported as intermediate coordinates, never as substitutes for root success. A local lemma can be mathematically correct while contributing nothing to the root. The gluing/interface metrics explicitly measure this failure mode.

The strongest capability claims will therefore rely on auxiliary task distributions with externally checkable or held-out outcomes in addition to the live open-problem observatory.

15.3 The underlying model can be the real cause

An agent may solve a task because the language model already knew how. This is the motivation for the four-arm design in Section 11. If `MODEL_ONLY` performs as well as `RAKL_LEARNING` at matched validity and resources, the data do not support a Orion capability benefit. If learning and sham-memory arms are similar, the content-specific memory hypothesis is not supported. If `RAKL_RESET` accounts for the gain, the effect belongs to static workflow scaffolding rather than continual experience.

15.4 Extra compute and context can masquerade as intelligence

Orion performs retrieval, preprocessing, validation and memory compilation. An unbudgeted comparison against a shorter direct prompt would be unfair. All arms therefore share a preregistered resource ceiling, and actual usage is reported. The sham-memory arm additionally controls, imperfectly, for the possibility that merely adding more tokens or structured context improves performance.

No sham is perfect. Irrelevant memory may distract more than an empty context, while carefully chosen near-miss memory may create a harder condition. We will therefore freeze sham construction rules before task outcomes and report more than one sham stratum when sufficient budget is available.

15.5 Prompt and workflow differences are part of the intervention

A full Orion-versus-model-only comparison necessarily changes instructions. It estimates the effect of the complete architecture, not a single isolated variable. The reset-versus-learning comparison is cleaner for experience because both arms use the same Orion workflow. Component ablations then localize specific mechanisms. We will not interpret the total Orion lift as a pure memory effect.

15.6 Model-service drift and non-determinism

Hosted model names can conceal backend changes, and repeated samples can differ even under nominally fixed settings. Exact model revision is recorded when available. Strong effects should be replicated across seeds/sampling configurations and, where the claim is intended to generalize, across more than one model family or revision. Failure to reproduce under a changed hosted model does not by itself distinguish framework failure from model drift, but it weakens claims of model-independent benefit.

15.7 Temporal dependence and pseudo-replication

Development episodes are not independent because later states contain earlier experience. Treating every cycle as an IID sample would underestimate uncertainty. The task is the unit for paired benchmark comparisons; fresh-transfer tasks all start from the same frozen learned state. Longitudinal curves remain descriptive unless the dependence structure is explicitly modelled. Multiple episodes generated from the same source or failure lineage are not counted as independent transfer validations.

15.8 Concurrent agents can contaminate lanes

Several research agents write to a shared application repository. One lane can therefore encounter artifacts produced by another lane between its start and completion. Exact base/head identities and fibre snapshots are required to establish what was available at decision time. A fresh-transfer benchmark uses frozen immutable states rather than a moving shared `main`. Uncontrolled cross-lane writes invalidate a benchmark packet but remain legitimate observational case evidence.

15.9 Telemetry changes the behavior being observed

Requiring agents to record failures, retrieved memory and saturation may itself improve or constrain behavior. This is not a defect if the telemetry is part of Orion, but it matters for interpretation. The pre-telemetry archive and reset/model-only arms provide partial controls. If Paper 5 claims that observation alone reveals natural model behavior, that claim would be too strong. We study the behavior of agents operating inside a specified research architecture.

15.10 Semantic novelty labels can be wrong

The seven-axis growth vector requires identity and semantic-retention judgments. Raw counts are objective relative to the stored data; retained novelty can depend on imperfect canonicalization. We therefore preserve the raw object stream, version novelty criteria, permit later supersession and avoid using retained novelty as a sole promotion metric. Periodic external/manual audits of sampled novelty assignments are desirable.

15.11 Residual contraction can be gamed

A framework can make the blocker set smaller by hiding or coarsening obligations. This would create artificial progress. Residual transformations therefore preserve evidence and are cross-checked against root contracts and gluing coverage. A step that exposes new independent blockers is not penalized merely because the count increases. Hostile tests should include obstruction renaming and deliberate omission of an interface.

15.12 Retrieval recall requires a bound universe

A claim that no relevant memory was found is not measurable against an unbounded repository narrative. The XM003 case exposed exactly this problem. Retrieval recall is therefore reported only relative to a content-identified index/universe and later relevance adjudication. A future framework-level coverage receipt is an explicit engineering requirement rather than an assumed property.

15.13 Same-session expert cells are not independent reviewers

The live research lanes use role-separated expert passes for adversarial coverage. These roles share model context and cannot be counted as independent peer review. The Paper 5 drafting/review loop in the present session has the same limitation. Nature-style concern taxonomies can improve coverage, but same-session internal review is labelled as such. Strong external-review claims require genuinely separate people/processes and evidence lineage where relevant.

15.14 Researcher and author intervention

The human operator chooses high-level goals, can change scheduled-agent prompts, may override merges and is co-designing the framework while observing it. The system is therefore not an experiment in fully autonomous AI development. Operator interventions are timestamped as part of the method history. Paper 5's claim concerns evidence-governed human/AI recursive research infrastructure, not human-free self-evolution.

15.15 The paper is being written while the system evolves

A live repository creates a moving-target risk. Every reported result therefore needs an explicit evidence cutoff and exact subject. Later findings can be added as new versions of the manuscript, but they cannot be silently backdated into the original evaluation. The present draft intentionally contains placeholders for prospective metrics rather than estimating them from incomplete future data.

15.16 Publication and confirmation bias

A self-improvement project naturally prefers stories of successful evolution. We counter this by preserving rejected variants, baseline-only wins, invalid transfers, failed consolidations, blocked promotions and operational overrides. Paper 5 should publish a complete registered evaluation packet, not only the tasks on which Orion helps. If the result is null or negative, the framework and paper claims must narrow accordingly.

15.17 A registered hard gate can be structurally unable to fail

This threat is not hypothetical for this paper; it was realized and measured in our own evidence apparatus. The first structured experience-to-action harness assigned gold and prediction from the same pure function, so four of its six registered gate conditions passed by identity, with thresholds registered exactly at ceilings the metrics could not leave. A black-box probe battery (perturbing evidence and observing whether each condition's verdict can move) detected the defect, a governance narrowing preserved the original receipt verbatim while withdrawing the four accuracy-shaped claims, and the repaired independent-oracle gate was audited per condition—never on the conjunction, which can hide dead legs behind one live one—before any confirmatory use, with a re-planted self-grading control reproducing the defect signature under identical probes. The exported discipline is a falsifier for every future gate in this programme: a gate condition that no evidence perturbation can flip supports no confirmatory claim, and conventional power analysis does not detect this failure mode because a constant arm produces confident intervals.

15.18 Security and evaluator independence are assumptions, not proofs

Content addressing and protected paths do not protect against a compromised repository administrator, CI service, dependency or assurance operator. The no-self-promotion proposition is conditional on these boundaries. Threats outside the trusted computing base are reported rather than treated as solved by hashes.

15.19 Decisive result branches

The prospective programme is designed so that important hypotheses can fail cleanly.

H1: accumulated Orion experience improves fresh research. Supported only if `RAKL_LEARNING` shows a material preregistered gain over `RAKL_RESET` on fresh transfer without a blocking validity/resource regression. Refuted for the tested distribution if the effect is materially negative. Inconclusive if confidence/coverage is insufficient.

H2: learned memory content, not extra context, causes part of the gain. Supported only if learning materially outperforms preregistered sham-memory controls. If sham and learned memory are equivalent, the content-specific claim is unsupported even if both outperform reset.

H3: Orion helps beyond the base LLM. Supported in scope only if total matched lift over `MODEL_ONLY` is positive on the registered primary QoIs or if a separately preregistered epistemic-safety benefit is material. A substantial excess of `BASELINE_ONLY_SUCCESS` tasks is evidence against the claim.

H4: experience reduces repeated structural failure. Supported if fresh-task repeated-failure rate falls without increasing false-transfer or invalid-authority rates. If the framework simply avoids old failures by making different but equally serious failures, the hypothesis is not supported.

H5: framework evolution is more than benchmark overfit. Supported for a particular parent–child edge only after a preregistered development gain transfers to a fresh, protected, lineage-independent assurance packet under matched resources. Development-only improvement is explicitly classified as local improvement rather than evolution evidence.

H6: Orion's operator/representation atlas approaches a high zero-invention regime. This Orion-triviality hypothesis remains open. A growing share of verified tasks solved with only preexisting resources would support it; a persistent need for new representations/operators/ontology across fresh tasks would refute or bound it.

These result branches are part of the method. They prevent future manuscript revisions from defining "success" after observing whichever metrics happen to improve.

16 An evidence-triggered roadmap from Orion 3.0 to later challengers

A roadmap can itself become a source of confirmation bias if version numbers are assigned to ideas before evaluation. We therefore distinguish *candidate sequence* from *released method sequence*. A label such as 3.1 is earned only after the corresponding challenger passes its governed process. If the first candidate fails, it remains a rejected branch and a different challenger may eventually become 3.1.

16.1 3.0.x: measurement and authority hardening without claiming a new research method

Several immediate changes are instrumentation or implementation hardening rather than intended research-policy innovations. Suitable patch candidates include:

- read-only state/process metrology and the four-arm attribution data structures introduced on the Paper 5 branch;
- content-bound replacement for the v3 archive's bare governance Boolean while preserving the existing constitutional requirement that promotion needs governance approval;
- content resolution of verification/review/proof IDs before authority-bearing use;
- certificate-backed local-section verification on gluing authority paths;
- promotion-grade binding of benchmark state/output identities to exact task/evaluator/artifact subjects;
- public API and CI stabilization.

These repairs should improve observability and enforcement of already stated principles. Unless they intentionally change search behavior, they do not by themselves support a new method-generation claim.

16.2 First Class-B candidate: coverage-bound research memory

The strongest currently observed cross-problem process defect is the XM002 retrieval miss: a synthesis stated that a Hodge reuse remained absent even though the exact base already contained the corresponding ResearchMemoryReview and reuse artifacts. Framework issue `RAKL#119` therefore proposes a content-bound cross-problem coverage receipt.

The candidate mechanism is straightforward. A memory query binds the exact problem/lane universe, index/manifests and structural query coordinates before making completeness-like statements. The primary prediction is lower missed-relevant-memory rate without requiring full-repository loading. Secondary predictions include lower duplicate research and better cross-domain tool reuse. Main risks are context/compute overhead and false confidence in an incomplete index.

The cheapest discriminating benchmark uses frozen repository snapshots with planted relevant and near-miss memories across several lanes, comparing the current query path with the coverage-bound challenger at matched retrieval budget. A successful development result still requires fresh surface-shifted tasks before the candidate can become a new method version.

16.3 Root-coordinate preservation audit

Several live case-study episodes exhibit a shared abstraction: an appealing local or surrogate quantity does not preserve the root-critical coordinate. Examples include raw closure mass versus cover complexity, positive norm versus Li-coefficient faithfulness, augmentation order versus complex analytic order, finite-cutoff spectral information versus continuum physical gap, and detector visibility versus actual geometric obstruction vanishing.

If the pattern survives enough independent episodes, a Class-B challenger can require an explicit `RootCoordinatePreservationAudit` before a surrogate becomes a major route. The intervention asks for the exact map from surrogate to root coordinate, non-preservation examples, cheapest hostile control and remaining gluing obligations. Its primary metric is reduction in root-coordinate false-progress/candidate-generation rate. Its main risk is excessive conservatism that blocks useful exploratory representations before a bridge can be discovered.

16.4 Gluing-aware search policy

Another recurring pattern is local mathematical success with an unresolved global interface. Yang–Mills provides an abstract spectral lemma whose target-specific OS/density/continuum binding remains open; Navier–Stokes repeatedly exposes far-field, compactness or limit-passage interfaces after valid local estimates. V3 represents gluing obstructions, but a later challenger could make interface risk an explicit routing coordinate.

The predicted benefit is earlier detection of "correct but non-closing" local work and higher residual contraction per cycle. The hostile control is a problem where local decomposition is sufficient and extra gluing machinery only adds overhead. The candidate should fail if it systematically diverts effort from locally solvable tasks.

16.5 Experience-conditioned routing and strategy motifs

V3 already contains experience-conditioned operator scoring and motif induction. The correct next action is evaluation, not automatic elaboration. Once enough v3 telemetry accumulates, compare routing with and without experiential priors on repeated-family, transfer and hostile near-miss tasks. A successful result should reduce repeated failures and/or cost while maintaining false-transfer control. If gains disappear under sham memory or fresh transfer, the mechanism should remain local or be revised rather than promoted as a general method improvement.

16.6 Consolidation policy as an evolvable object

The v3 fast/slow split intentionally prevents every episode from being immediately rewritten into a lesson. Later data may support different consolidation schedules by failure type, task family or evidence density. This is a natural candidate for method learning because external research has demonstrated that forced continuous consolidation can degrade memory. Any challenger must preserve raw episodes and be compared against episodic-only and current gated-consolidation controls.

16.7 Portfolio and saturation policy

The long-horizon autonomous-research literature and our own saturation vector both suggest a failure mode in which an agent repeatedly hill-climbs within one method family after returns flatten. A future challenger could use measured axis-specific saturation and stable residuals to reallocate budget among exploitation, diversification, moonshots and meta-method search. The benchmark must include tasks where persistence within one family is actually correct so that diversification is not rewarded merely for being novel.

16.8 Meta-method invention

Only after relevant operator/path/meta-method axes are demonstrably saturated and stable residuals remain should the system search for a genuinely new research operator or ontology. The v3 invention gate operationalizes this principle. Paper 5 treats operator invention as a measured exception rather than the default explanation of progress. This creates an empirical test of the Orion-triviality hypothesis: how often do verified solutions require new primitive research structure rather than recomposition or transfer?

16.9 What later versions should publish

Each released method version should have a compact "model card" style evidence packet:

- exact parent and candidate Git/method identities;

- source pathologies or positive motifs motivating the change;
- preregistered mechanism and predicted metrics;
- matched development results and uncertainty;
- fresh protected assurance results;
- regressions and baseline-only wins;
- resource profile;
- state/schema migration and rollback status;
- evaluator/assurance identities and exposure history;
- final governance/promotion and active-main attestation.

Paper 5 can then become a longitudinal scientific record of the sequence

$$\mathrm{Orion}_{3.0} \to \mathrm{Orion}_{3.1} \to \mathrm{Orion}_{3.2} \to \cdots$$

without assuming in advance that every arrow points to a better system.

17 Discussion

The central claim of Method-Evolution Mechanics is deliberately narrower than "an agent that remembers becomes a better scientist." A long-running research process exposes typed longitudinal evidence about representations, retrieval, repeated failure, verification, route changes and method revisions. If that record is preserved rather than collapsed into free-form reflection, the research method itself becomes an auditable object. The resulting system may alter what it proposes or inspects next while scientific authority remains governed by a separate evidence path.

The evidence in this paper now separates three levels that are often conflated. First, the experience and method-state architecture is implemented: task episodes, lessons, failures, structural transformation memories, observability, method variants and protected governance objects are externally represented and content-bound. Second, the invention-last routing mechanics are experimentally and operationally hardened: structured `SEARCH`, `JUMP`, `GLUE`, `LIFT` and `CANNOT_CHECK` decisions survive stronger-parent, hostile-mutation and exact-head production tests, including typed negative history and conclusive-rejection fallthrough. Third, whether an LLM can infer and exploit these structures to improve fresh research outcomes remains a distinct causal question. The existing model ladder does not provide a capability-qualified operating point for that comparison, so the four-arm efficacy estimand remains unidentified rather than being imputed from the first two levels.

17.1 Failure and negative history are part of the method state

A failed attempt is useful only if the system preserves enough structure to avoid repeating the same unsupported transition. The routing campaign makes this concrete. A failed candidate is not equivalent to an absent candidate, and an unresolved candidate is not equivalent to a rejected one. Conclusive rejection therefore becomes a typed, content-bound transition that records candidate identity, revision, target, audit evidence and the negative reason. The same principle applies at composition level: rejecting one GLUE composition does not delete its component episodes because they may participate in another viable composition. This is a stronger form of "learning from failure" than storing a textual post-mortem: future control flow depends on a replayable negative certificate.

17.2 Why the four-arm causal benchmark remains necessary

A promotion-grade routing mechanic does not establish a promotion-grade learning effect. The underlying model may already solve the task; static scaffolding may explain a gain; persistent state may help without learned content being useful; or memory content may redirect search in a genuinely causal way. The registered MODEL_ONLY, RAKL_RESET, RAKL_SHAM_MEMORY and RAKL_LEARNING comparison exists to distinguish these possibilities. The current capability-floor result prevents an interpretable execution at the registered operating point. This is an empirical boundary, not a reason to relabel architectural success as model efficacy.

This separation also protects negative results. The 0.5B precursor and the exhausted 0.5B–7B qualification ladder remain first-class evidence. They show that a longitudinal state transition can be implemented correctly while the model still fails the task-level success gate. In that regime, a treatment contrast is not evidence for or against useful learning because the necessary model capability is absent.

17.3 Governed self-evolution is proposal authority, not scientific authority

The wider RSHEA/self-hosting stack turns observability, meta-controller decisions and method challengers into content-bound records, but preserves an explicit authority split. Evolution evidence cannot substitute for control authority; failed hard gates cannot be compensated by positive utility; shadow decisions do not act; governed interventions require separate sign-off; and runtime resumption is bound to the evaluated epoch and ledger. These properties make recursive method change experimentally inspectable without giving the method permission to certify its own scientific claims.

The trusted computing base is not eliminated. Repository access, CI, governance credentials and human administration remain external assumptions. The defensible statement is process separation: within the registered software and evidence boundaries, proposal generation, routing, learning and self-evolution do not mint theorem or scientific authority by themselves.

17.4 What the Millennium case study contributes

The six unsolved Millennium programmes are used as a hostile longitudinal observatory, not as evidence that autonomous agents are close to solving the problems. They expose exactly the situations in which method memory matters: local results that fail to connect to the root quantity, structurally tempting transfers whose preconditions do not survive, missed prior work, incompatible partial methods, repeated residuals and verification-induced route changes. A local lemma or negative obstruction can be valuable even when it does not solve the root. Conversely, an attractive representation with no faithful root bridge is a risk that the method state should remember.

17.5 Threats that remain open

Several scientifically important coordinates are intentionally outside the promotion described here. The generated routing validation operates on already-structured method state; natural-language obstruction extraction is not solved by it. The four-arm model-level efficacy study remains capability-limited. The retained-growth metrology has a mechanically auditable subset but broad semantic novelty has no independent human validation in the frozen record. Provider-level compute economics may also change the practical value of a safe route even when the routing decision itself is correct. These boundaries are carried into the claim-to-receipt matrix rather than hidden as generic "future work."

18 Conclusion

We introduced an evidence-governed formulation of research-method evolution in which longitudinal experience is external, typed, versioned and non-sovereign with respect to scientific authority. The current system records episodes, lessons, failures, structural transformations, method variants and governance state, and it treats the research trajectory as an empirical object rather than a narrative memory.

The paper now closes its core method-governance mechanics at a stronger level than the original architecture-only proposal, and it does so on audited instruments. The experience-to-action mechanic is held to a repaired gate whose gold is assigned post-freeze by an independently implemented oracle and whose eight evidence-dependent conditions are individually demonstrated capable of failing; under it the shipped mechanic is exactly conformant on a fresh 2,400-case panel, the strongest composite simple parent remains materially information-limited (0.433 ceiling), and all twelve planted candidate mutants are caught. The earlier 2,688-case routing validation is deliberately not cited: its harness is absent from the repository and the result carries `CANNOT_CHECK` status until it is reproduced or withdrawn. The production routing stack binds positive routes to the canonical audit, uses content-bound candidate rejection before SEARCH/JUMP fallthrough, uses exact composition rejection before GLUE-to-LIFT fallthrough, preserves negative history and blocks missing, stale or non-conclusive evidence. Exact-head routing, protected repository and artifact-contract gates are green for the final stacked subject.

The broader causal learning thesis is not claimed. The preregistered capability ladder did not produce an operating point at which the four-arm comparison could identify architecture, persistent-experience and learned-content effects. That lane therefore terminates here as capability-limited/unidentified for this release rather than being rescued by post-hoc model shopping.

The resulting scientific statement is consequently two-part: *method state can be made operationally evolvable under promotion-grade fail-closed routing and governance contracts; whether that method evolution improves fresh LLM research performance remains a separately registered causal question.* Preserving both parts—the promoted mechanics and the negative capability boundary—is the intended evidence discipline of the paper.

Code, data, materials and AI-use disclosure

A public research artifact accompanies this paper, with the mathematical application case-study maintained as a separate companion artifact. Research agents, coding agents and language models were used as proposal, implementation, literature-search and drafting tools. They are not represented as independent reviewers, human experts or authors. Public research traces are auditable decision records, not disclosures of hidden model chain-of-thought. The longitudinal mathematical programme studies unsolved problems and carries no claim that any Millennium Prize Problem has been solved unless a separate root certificate satisfies the applicable proof, dependency, novelty and independent-review gates. Prospective evaluation quantities that have not yet been measured are marked as preregistered targets or placeholders rather than reported as results.

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Case	Observable event	Process diagnosis	Framework hypothesis
Cross-problem XM003	a synthesis missed an already-present Hodge reuse of a registered cross-domain audit tool	retrieval coverage failure, not chronology failure	completeness/no-match claims need a bound coverage receipt
P-vs-NP XM004	raw source-native closure cardinality becomes maximal in a source-defined zero-cover-complexity regime	surrogate representation mass does not preserve root-critical cost	require a root-coordinate preservation audit before scoring a representation
RH analytic/spectral	positive or trace-like surrogate objects hide RH-equivalent faithfulness/endpoint obligations	representation and faithfulness defect	make bridge defect explicit and falsify the interface before theorem invention
BSD S001c1a	augmentation order contains local corrections and extra zeros unrelated to the desired complex rank coordinate	scalar representation is not root-faithful	rotate to corrected or richer graded objects rather than absorb excess vanishing by notation
Yang–Mills S1a1	an abstract dense-source spectral lemma is valid, but target OS source binding, uniformity, RG transport and continuum identification remain open	local theorem success with unresolved global interfaces	treat local-to-global gluing obligations as first-class residuals
Navier–Stokes B2a	a Type-II route yields an ancient Euler limit, making ordinary parabolic Navier–Stokes backward uniqueness inapplicable	unsafe method transfer across changed equation/interface	require source-to-target inheritance witnesses before reusing rigidity tools
Yang–Mills E1a1a0	primary metadata identified the correct Bałaban averaging source, but accessible text did not expose the exact nonlinear map/contract	source-completeness failure	missing source detail should block candidate generation rather than be reconstructed from memory
Orion merge incident	v3 was merged at operator request before CI cleanup; a later PR repaired CI and a benchmark float assertion	deployment occurred before assurance closure	deployment and method authority need separate states plus explicit operator-override records

Table 3: Initial qualitative cases. These are hypothesis-generating observations, not estimates of prevalence or causal treatment effects.

Metric	Definition	What it indicates
Repeated-failure rate R_F	failures whose structural signature was seen before, over all failures	whether the system actually learns from its own failures; target decreasing
Lineage coverage	derived canonical objects with valid source ancestry, over all derived objects	whether growth is grounded rather than orphaned
Unresolved-link / orphan rate	dangling references in the common substrate, over all links	silent decay of the knowledge base
Supersession-preservation rate	superseded records retained in history, over all supersessions	whether negative/old history is preserved, not overwritten
Invalid-promotion rate	promotions lacking a valid governed certificate, over all promotions	whether proposal/authority separation holds; target 0

Table 5: Method-health metrics an implementer can instrument on a long-running agent without the causal-attribution experiment. Reported as a profile; each measures process discipline, not task accuracy.